UNIVERSITI TEKNIKAL MALAYSIA MELAKA

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FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY

W O R K S H O P 1

R E P O R T

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Program :BAXI

Project Title : Community Service Request Management System

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EXECUTIVE SUMMARY

The Community Service Request Management System is developed to address common challenges in managing public service requests within a community. With the growing demand for services such as electricity maintenance, water supply, road repair, and custom services, manual and unstructured systems often lead to delays, poor tracking, and customer dissatisfaction. This project aims to improve efficiency, transparency, and service quality through a centralized system.

Many existing systems rely on manual processes that cause data inconsistency, slow response times, and difficulties in monitoring service requests. Customers face challenges in tracking their requests, while administrators struggle to manage records and generate useful reports. These limitations reduce operational effectiveness and overall service performance.

The objective of this project is to design and implement a console-based system using C++ that allows customers to submit and track service requests, while enabling administrators to manage services, update request statuses, and handle customer information efficiently. The system automates key processes, reduces manual errors, and improves response time.

The scope of the project includes user management, service selection, request submission, payment processing, and request tracking with database support. This project demonstrates the practical application of programming and database concepts, and its significance lies in enhancing community service management through improved efficiency, reliability, and customer satisfaction.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The Community Service Request Management System is a modern solution designed to improve the management of public service requests within a community. In an environment where efficiency, organization, and timely service delivery are essential, there is a strong need for effective systems that enhance coordination between service providers and customers while increasing overall satisfaction.

Traditional methods of managing community services often depend on manual processes, which result in delays, data inconsistency, and operational errors. Customers face difficulties in submitting service requests and tracking their status, while service providers struggle with organizing records, scheduling services, and managing large volumes of requests efficiently.

This system is developed using C++ and provides an integrated platform for service request submission, service selection, payment processing, and real-time request tracking. With a user-friendly interface, customers can easily manage their service requests, while administrators benefit from automated workflows that reduce manual workload, improve data accuracy, and enhance overall service management efficiency.

1.2 Problem Statement

Inefficient Scheduling and Service Management: Manual booking and service management processes often lead to delays, scheduling conflicts, and poor coordination between customers and service providers, resulting in reduced service efficiency and customer dissatisfaction.

Inaccurate and Inefficient Record Keeping: The use of manual record systems increases the risk of data loss, duplication, and human errors, making it difficult to manage customer information, service history, and payments accurately. Inefficient Scheduling and Service Management: Manual booking and service management processes often lead to delays, scheduling conflicts, and poor coordination between customers and service providers, resulting in reduced service efficiency and customer dissatisfaction.

Inaccurate and Inefficient Record Keeping: The use of manual record systems increases the risk of data loss, duplication, and human errors, making it difficult to manage customer information, service history, and payments accurately.

1.3 Objective (s) of the project

This project embarks on the following objectives:

1. **To implement four (4) modular functionalities with CRUDS (Create, Read, Update, Delete, Search) database management for the Community Service Request Management System:**
    1. User Management Module
    2. Community Request Management Module
    3. Service Management Module
    4. Report Management Module
2. **To implement request tracking and status monitoring as the complex calculation required.**
3. **To develop summary and analytical reports for the Community Service Request Management System.**

1.4 Scope

i. Admin:

- Register (initial admin account setup)
- Log in
- Insert new Customer and Service
- View list of Customers, Community Requests and Services
- Update Customer, Community Request status and Service details
- Update Admin information
- Delete Customer, Community Request and Service
- Search for Customer, Community Request and Service by keyword
- View reports for monthly, annual, custom date and data visualization.

ii. Customer:

- Register
- log in
- Submit new community service request
- View list of own community requests
- Delete own community service request
- Track community request status
- Update own personal information
- View and print receipt .

1.4.1 Modules to be developed

i. Registration Module

- New customers and administrators are required to register their personal details to create an account and access the system

ii. Log in Module

- All users (Admin and Customer) must log in to the system using a registered user ID and password.

iii. Viewing Module

- Admin can view the list of Customers, Community Requests, and Services available in the system
- Customer can view the list of own community service requests and their details
- Admin Viewing Report

iv. Inserting Module

- Customer can submit new community service requests into the system
- Admin can insert new Services and Customer into the system

v. Updating Module

- Admin can update customer information, community request status, and service details
- Admin and Customer can update their own personal information.

vi. Deleting Module

- Customer can delete own community service request
- Admin can delete customer accounts, community requests, and services

vii. Report Viewing Module

- Admin can view reports based on monthly, annual, and custom date ranges with basic data visualization.

viii. Viewing Receipt Module

- Customers can view and print the receipt

1.4.2 Target User

Target User: Community Service Administrator

Role: Admin

Modules can be accessed:

i. **Inserting Module**
   - Admin can insert new customer accounts and community services into the system

ii. **Updating Module**
   - Admin can update customer information, community service request status, and service details
   - Admin can update own personal information

iii. **Deleting Module**
   - Admin can delete customer accounts, community service requests, and services from the system

iv. **Searching Module**
   - Admin can search for customers, community service requests, and services by keyword

v. **Viewing Module**
   - Admin can view the list of customers, community service requests, and services
   - Admin can view analytical reports based on monthly, annual, and custom date ranges with data visualization

Target User: Community Member
Role: Customer

Modules can be accessed:

i. Inserting Module
   - Customer can submit new community service requests into the system

ii. Deleting Module
   - Customer can delete own community service requests

iii. Viewing Module
   - Customer can view the list of own community service requests and track their status

iv. Updating Module
   - Customer can update own personal information

1.5 Project Significance

The Community Service Request Management System is designed to provide significant benefits and a positive impact on both community members and service administrators. The key benefits of the system include:

i. **Improved Community Engagement and Satisfaction**

- By providing a user-friendly platform for submitting and tracking community service requests, community members can easily communicate their needs and receive timely updates on request status. This improves transparency and enhances overall user satisfaction.

ii. Operational Efficiency

- The automation of request handling and administrative processes reduces manual work and minimizes errors. Administrators can manage requests more efficiently, prioritize tasks effectively, and ensure smoother service operations, leading to improved productivity.

iii. Accurate Data Management

- The centralized data storage ensures that all community service requests and user information are accurate and up to date. This allows administrators to analyze request patterns, generate reliable reports, and support informed decision-making.

iv. Scalability and Adaptability

- The system is designed to support future expansion, allowing additional services, users, or system features to be integrated easily. This ensures that the system remains flexible and relevant as community service needs evolve.

1.6 Gantt Chart of Project Activities

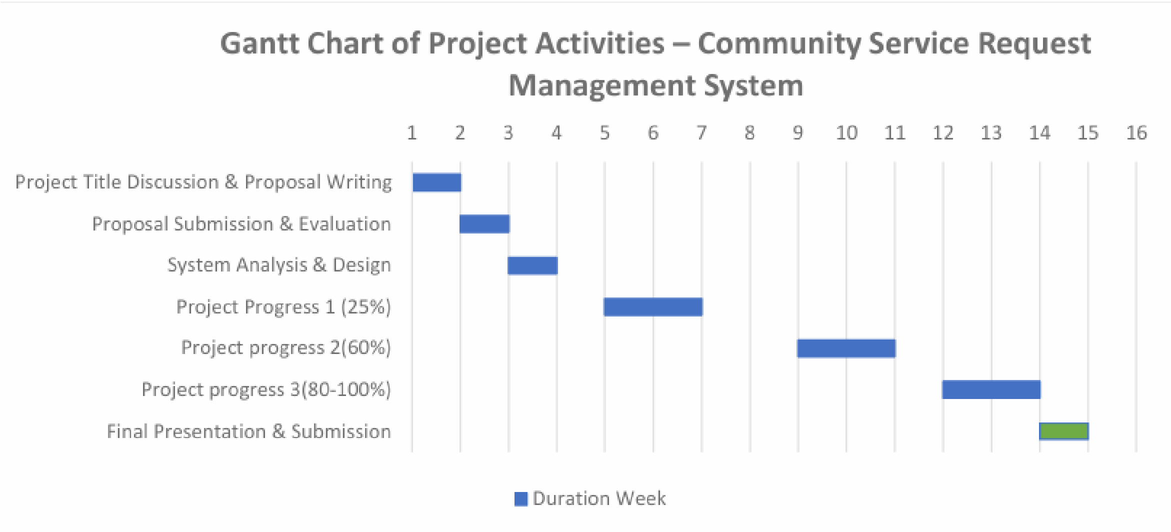

Figure1.1: Gantt Chart of Project Activities

CHAPTER 2: ANALYSIS OF PROBLEM

2.1 Problem Decomposition Description

- Sub-Problems and Steps to Solve Each

Sub-Problem 1: Inefficient Handling of Community Service Requests

Steps to Solve:

i. Analyze the current method of handling community service requests (manual submission, verbal communication).
ii. Identify common issues such as lost requests, delays, and lack of proper tracking.
iii. Design an online request submission interface for community members.
iv. Implement a centralized system to manage and organize all submitted requests.

Sub-Problem 2: Lack of Request Status Tracking

Steps to Solve:

i. Identify the different request statuses (pending, in progress, completed).
ii. Design a mechanism for administrators to update request status.
iii. Allow customers to view and track the status of their submitted requests in real time.

Sub-Problem 3: Poor Data Management and Record Organization

Steps to Solve:

i. Evaluate current data storage methods that lead to inconsistent or inaccurate records.
ii. Design a structured database to store user information, service details, and requests securely.
iii. Implement data validation to reduce errors during data entry.
iv. Ensure administrators can manage and maintain records efficiently.

Sub-Problem 4: Difficulty in Monitoring and Decision Making

Steps to Solve:

i. Identify key information needed for monitoring community service performance.
ii. Develop reporting features to summarize requests based on time periods.
iii. Generate analytical reports to support administrative decision-making.

2.2 Structured Chart

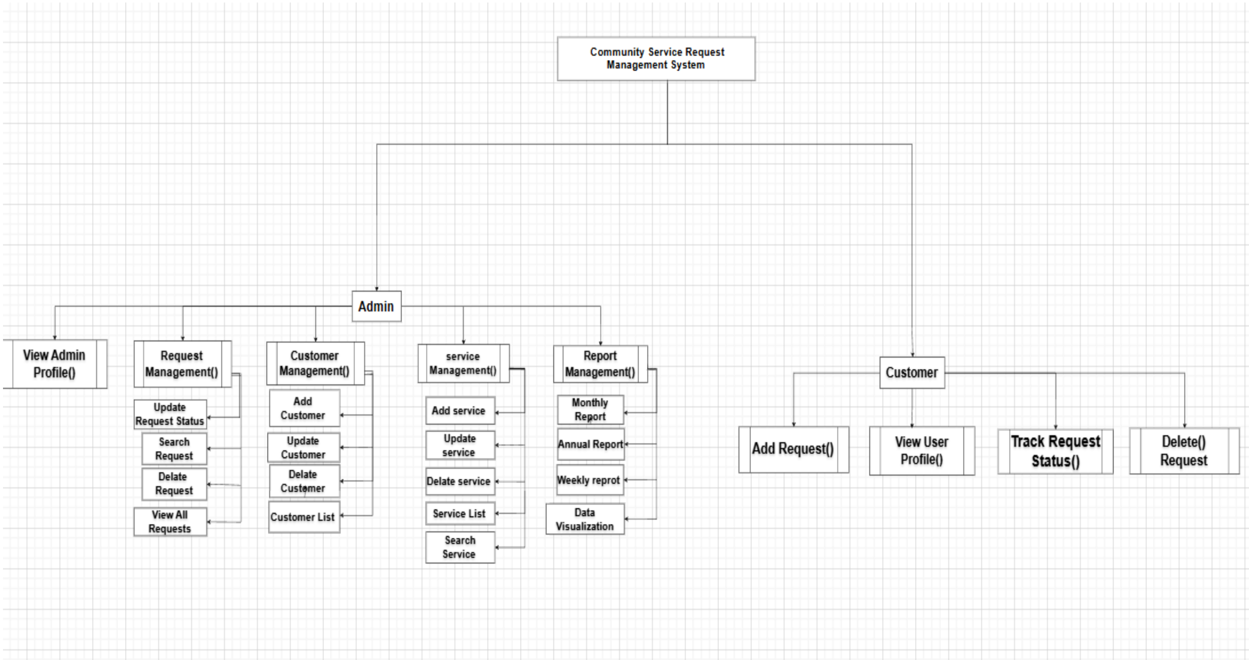

Figure 2.01: Structured Chart of the Community Service Request Management System

CHAPTER 3: DESIGN

3.1 Business Rules

Registration Rules

i. Customers must register by providing valid personal information such as name and email address.

ii. The system automatically generates a **unique user ID** for each registered customer.

iii. The email address must be **unique**, and duplicate email registration is not allowed.

iv. All registration details must be validated before the account is created.

v. Registration will be rejected if duplicate or invalid information is detected.

Login Rules

i. Users must log in using the system-generated user ID and their password.

ii. User passwords are encrypted (hashed) before being stored in the database.

iii. During login, the entered password is compared with the encrypted password stored in the system.

iv. Users are allowed a maximum of three (3) failed login attempts.

v. If the maximum number of failed attempts is exceeded, the system will temporarily deny access.

Administrator Validation Rules

i. Administrator access is restricted and requires additional verification.

ii. Only users with a valid admin verification code are granted administrator privileges.

iii. Customers are not allowed to access any administrative functions.

Data Validation and Security Rules

i. All user inputs must be validated to prevent empty or invalid data entries.

ii. The system must prevent duplicate user IDs and email addresses.

iii. Passwords must meet minimum security requirements.
iv. All failed login attempts must be recorded for security monitoring purposes.

User Management Rules

i. Only registered users are allowed to access the system
ii. Each user must log in using a unique ID and password
iii. Users are assigned roles (Admin or Customer), and access rights are granted based on the assigned role.
iv. Administrators are allowed to add, update, view, and delete customer accounts.
v. Customers are allowed to update only their own profile information.

Community Service Request Management Rules

i. Only customers can submit new community service requests.
ii. Each request must contain valid and complete information before submission.
iii. Customers are allowed to delete their requests only if the request status is pending.
iv. Administrators are responsible for approving requests and updating request status.
v. Request statuses must follow a predefined sequence (Pending → In Progress → Completed).

Service Management Rules

i. Only administrators are allowed to manage services in the system.
ii. Administrators can add, update, delete, and search for services.
iii. A service must exist in the system before it can be assigned to a request.
iv. Cannot delete service linked to completed request but for the pending and progress request have to manage then delete.

Report Management Rule

i. Only administrators are allowed to access system reports.
ii. Reports must be generated based on selected time periods (weekly, monthly or annually).
iii. The system must ensure that reports reflect accurate and up-to-date data.
iv. Data visualization must be generated automatically based on report data.

3.2 Flowchart

Main page:

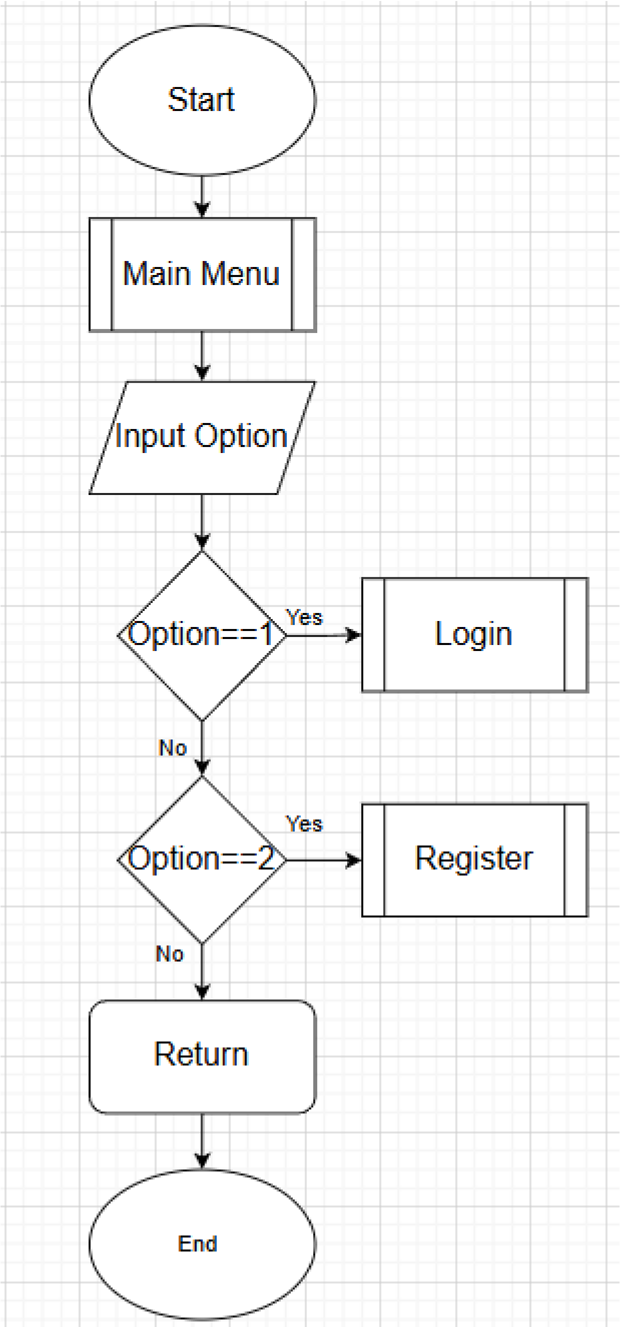

Figure 3.1: Main menu Module

Login Page:

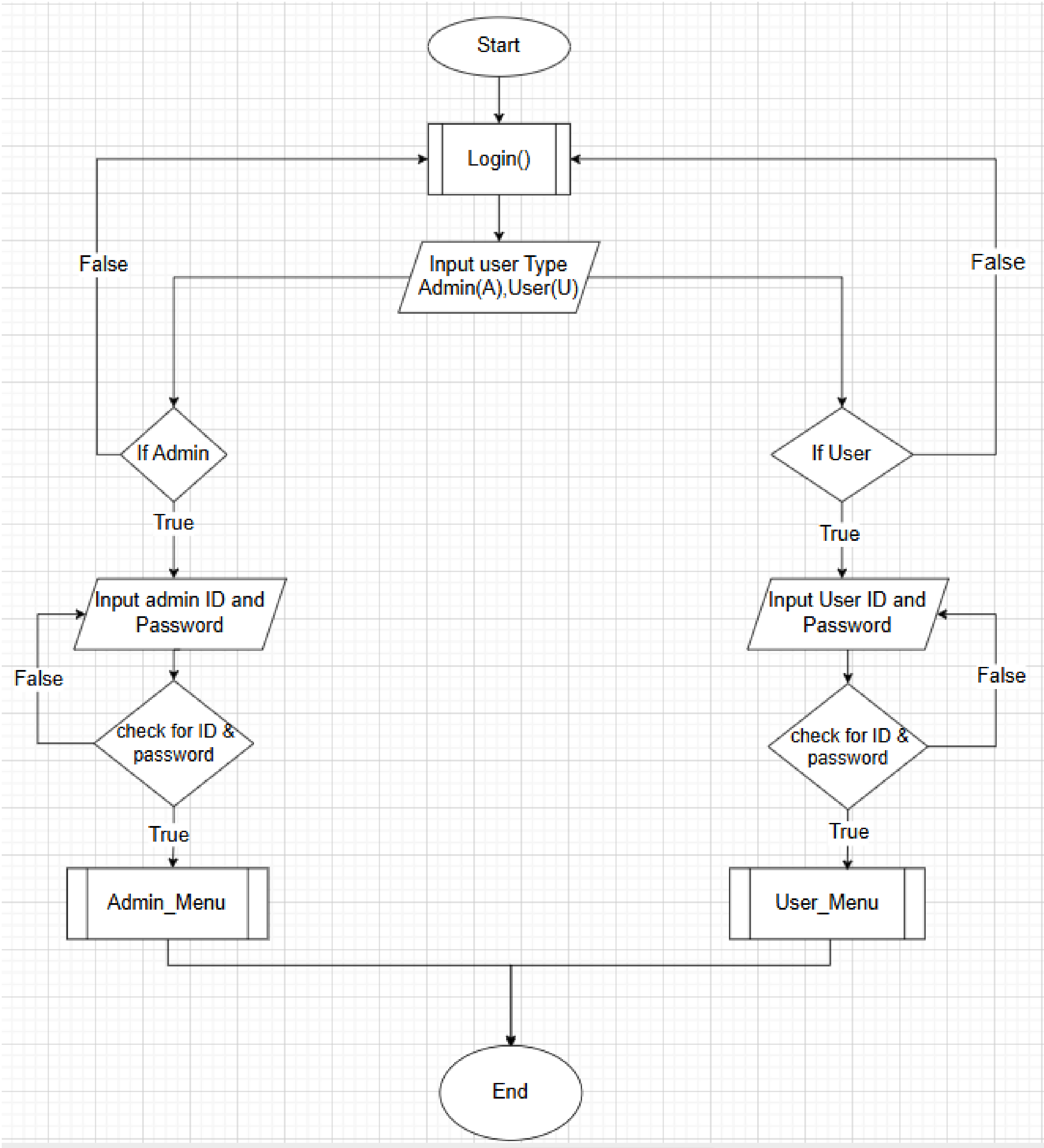

Figure3.2: Login Module

Register:

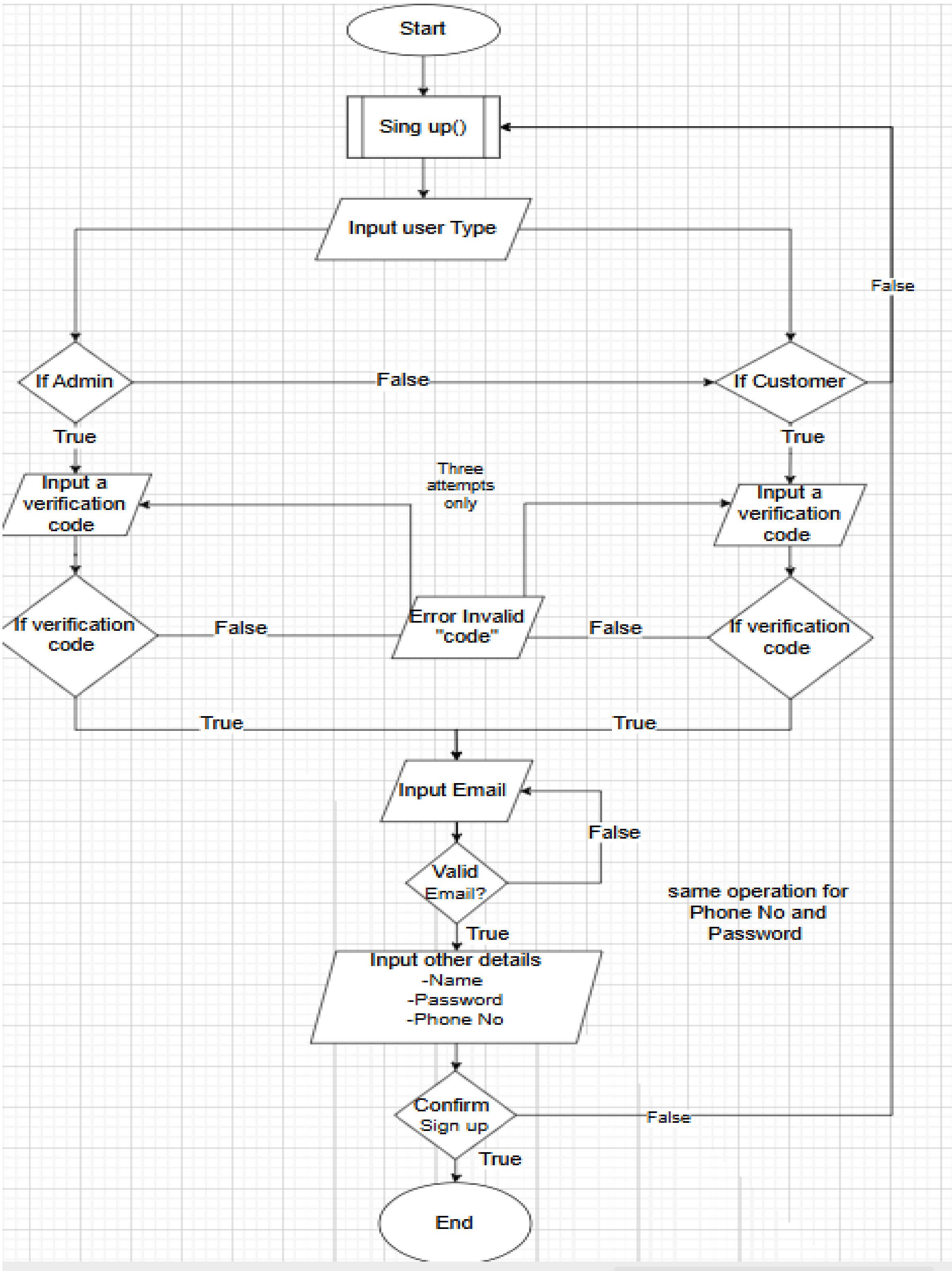

Figure3.3: Sign up Module

Admin Menu:

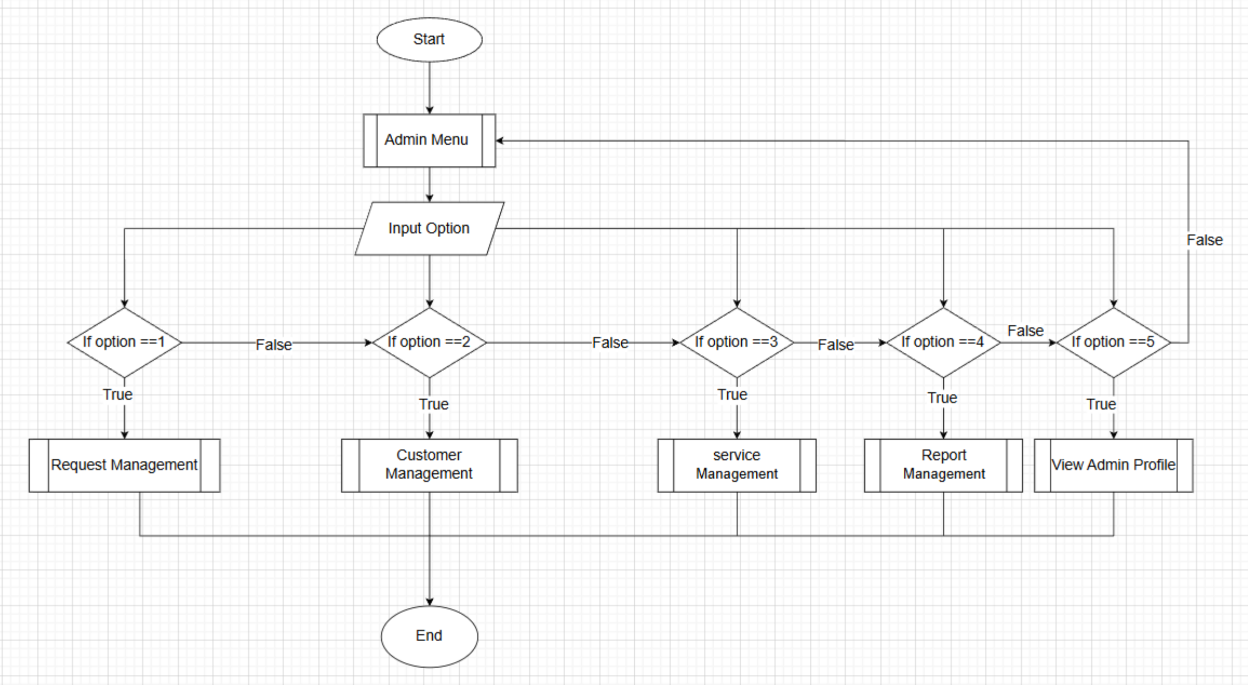

Figure3.4 : Admin Menu Module

Request Management:

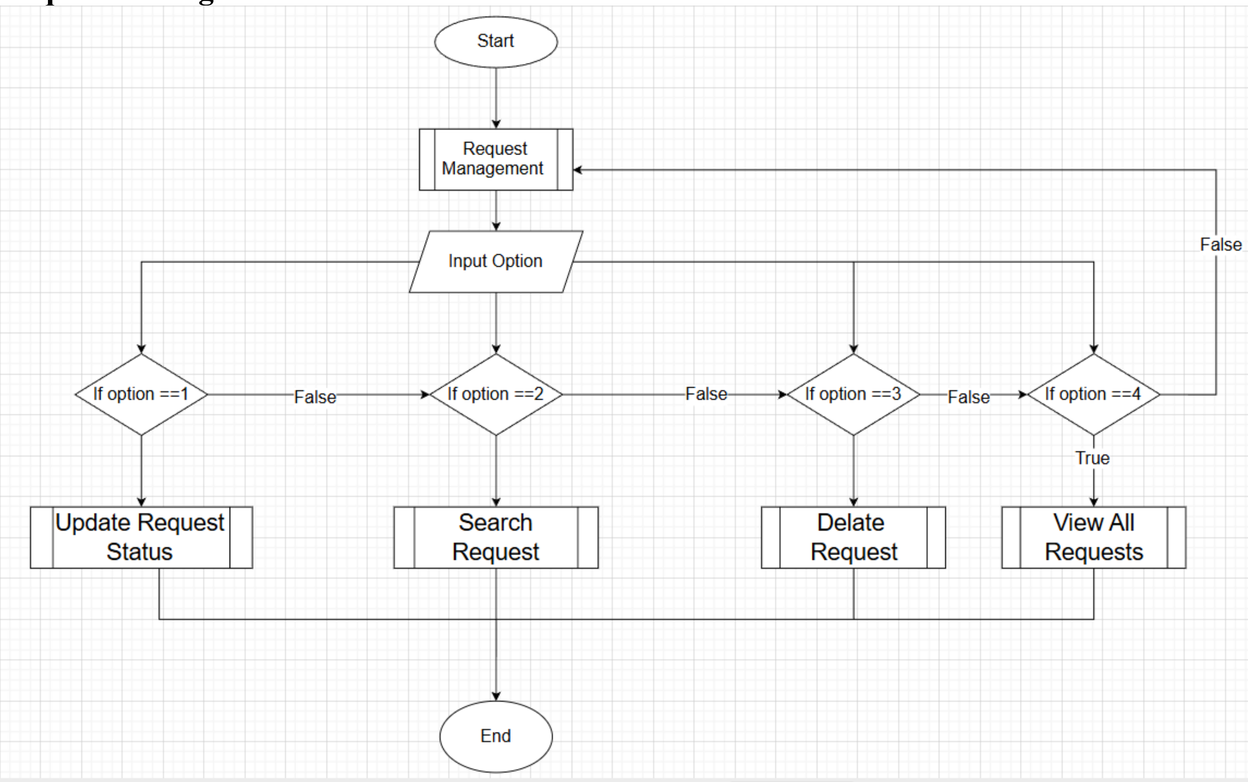

Figure3.5 : Admin Request Management

Update Request:

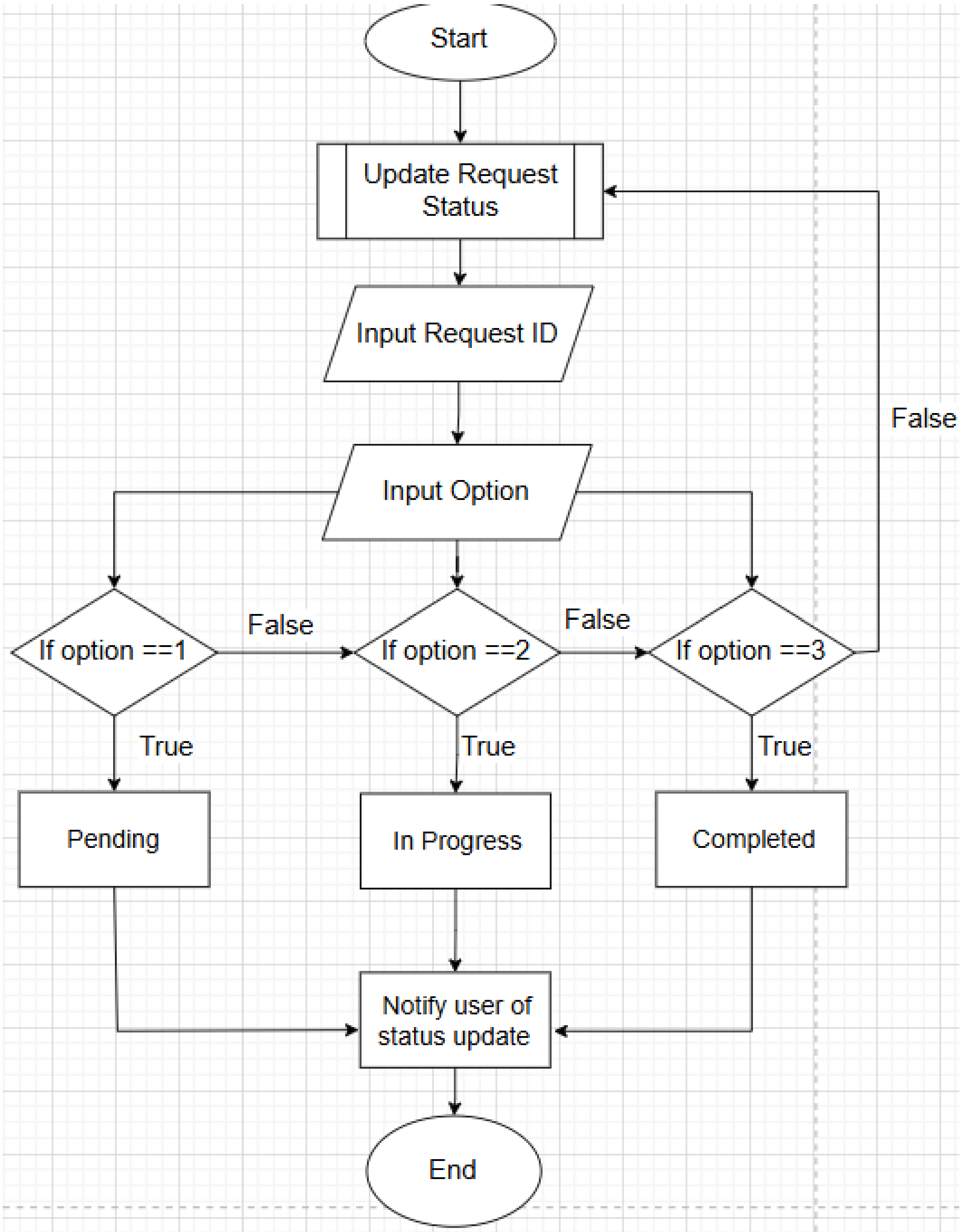

Figure3.6 : Admin Request management

Delete Request:

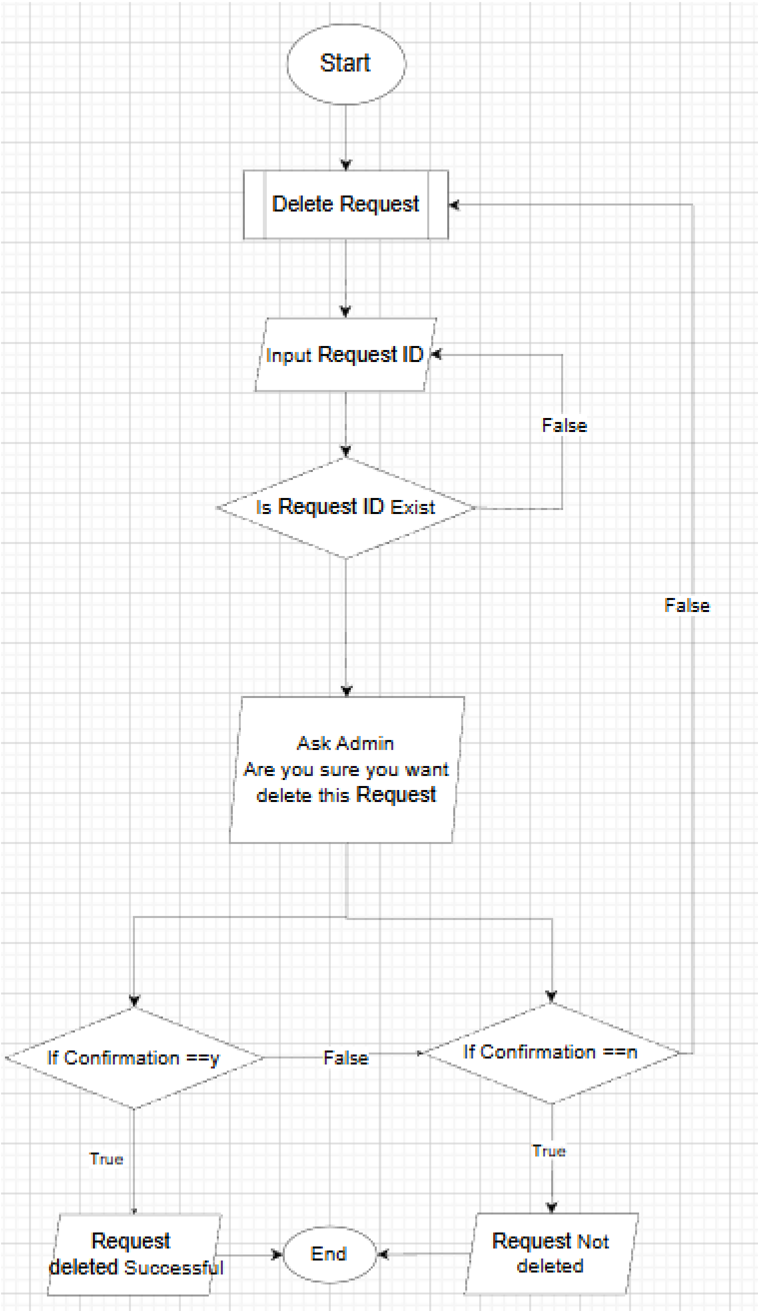

Figure 3.7 : Admin Delete Request

Search Request:

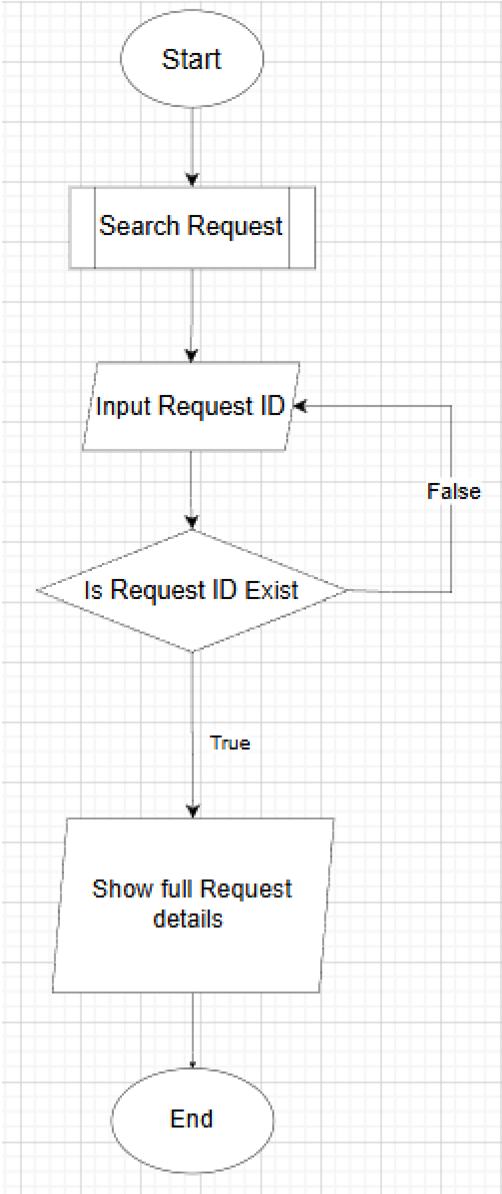

Figure 3.8 : Admin Search Request

Service Management:

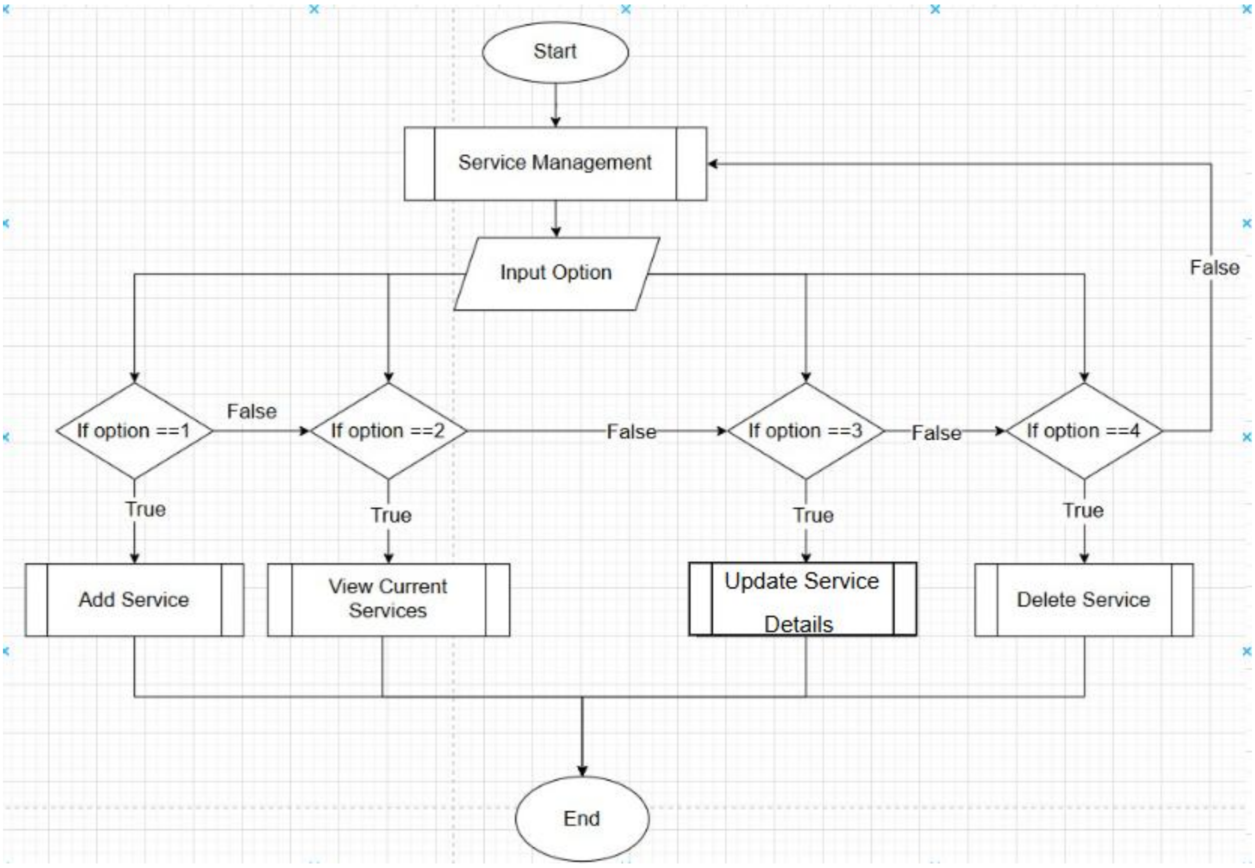

Figure 3.9 : Admin Service Management

Add Service:

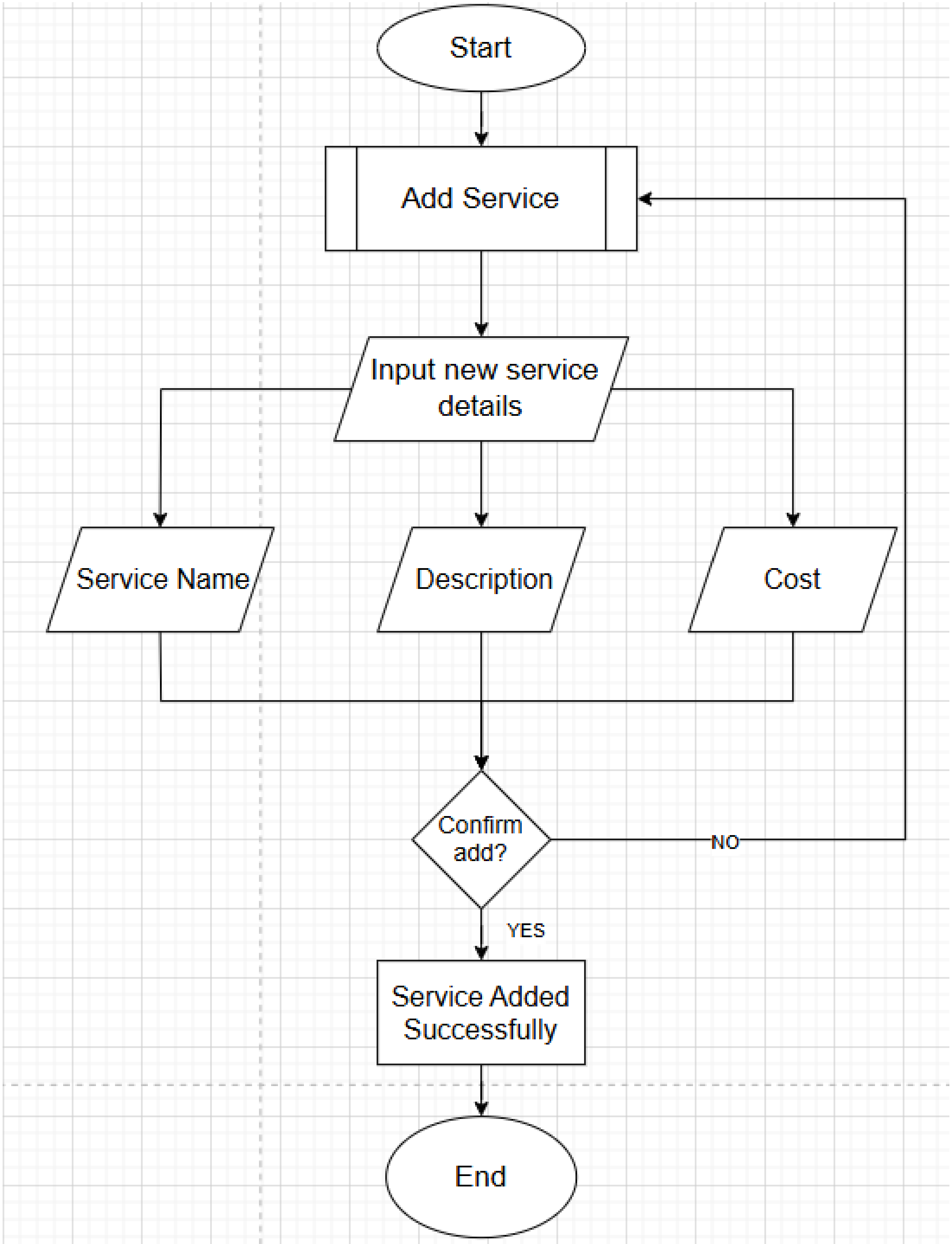

Figure 3.10 : Admin Add Service

Update Service:

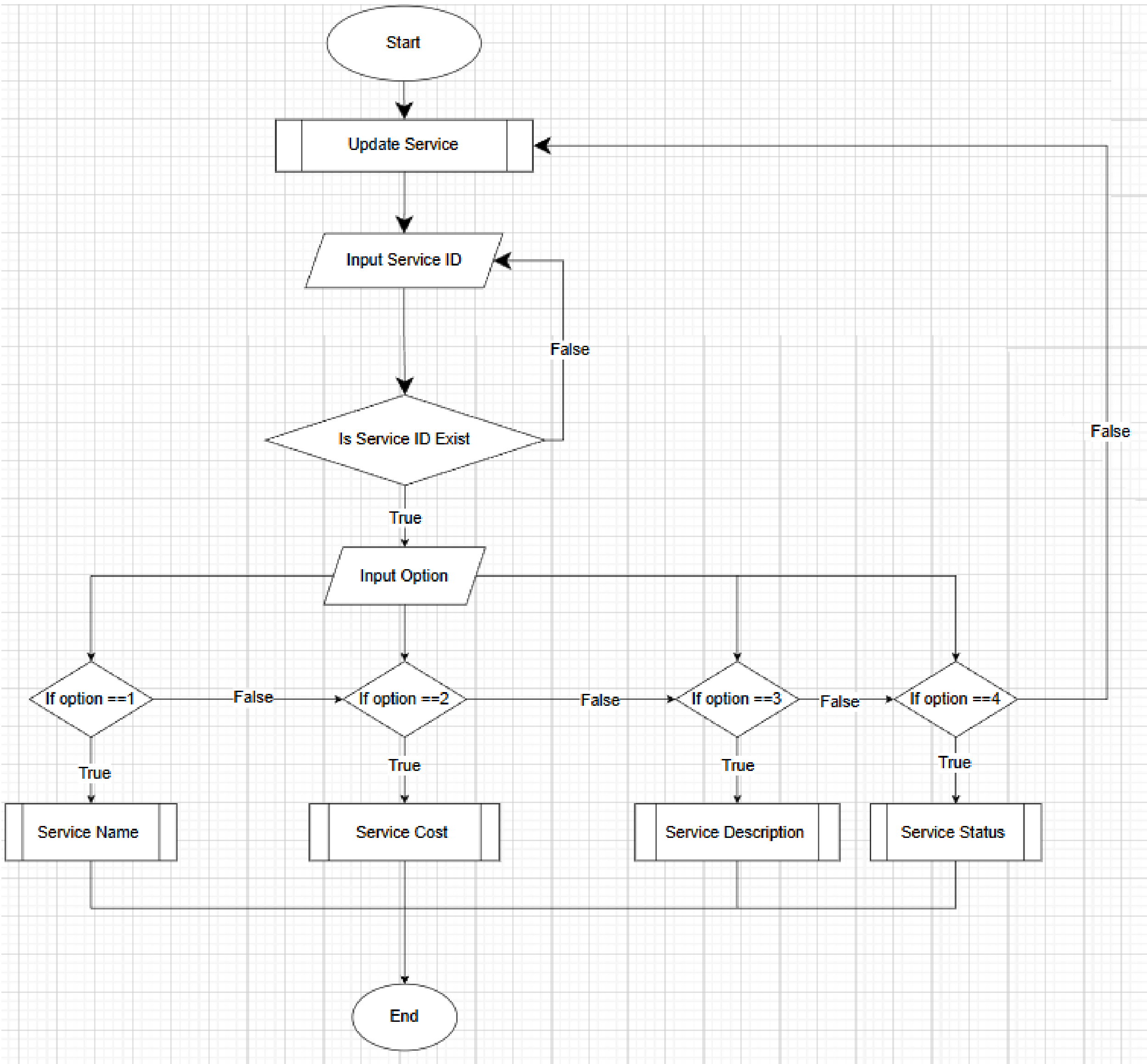

Figure 3.11 : Admin Update Service Details

Delete Service:

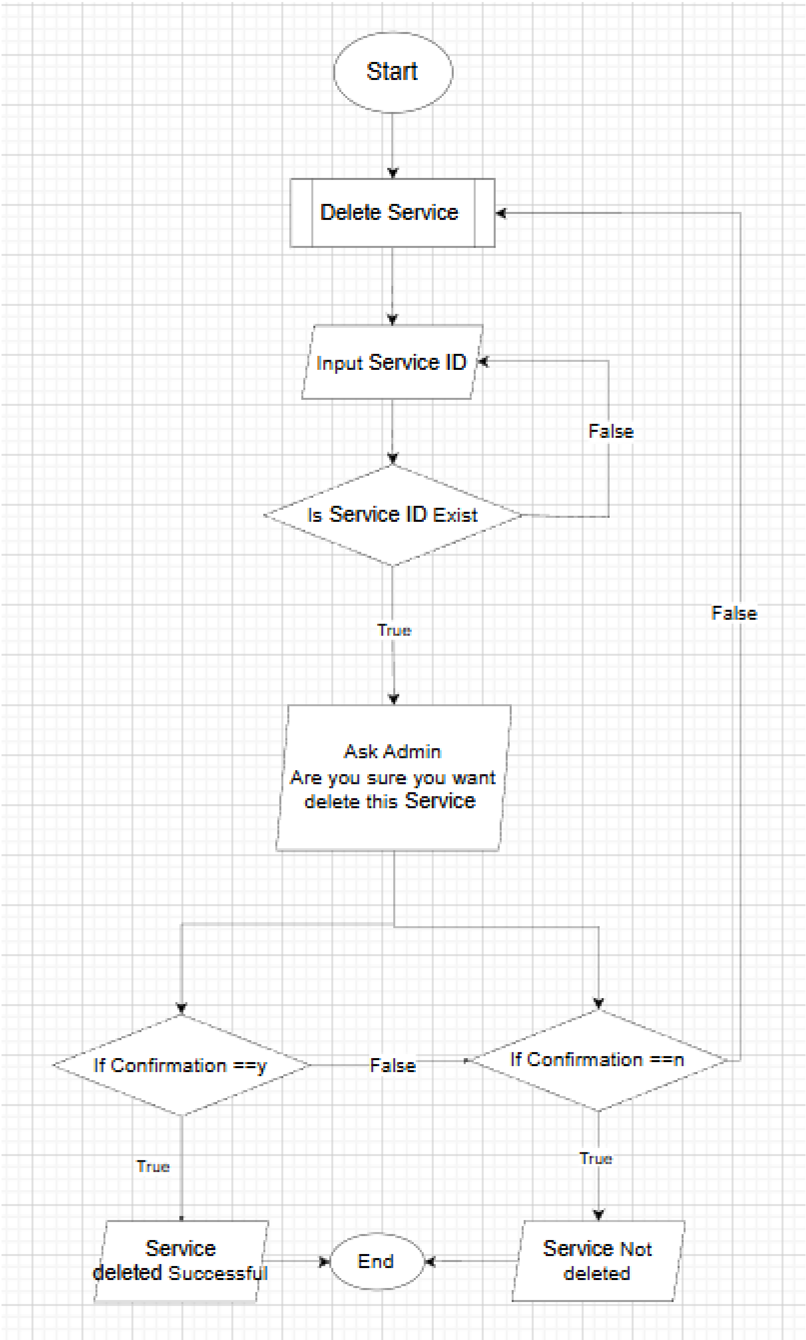

Figure 3.12 : Admin Delete Service

Report Management:

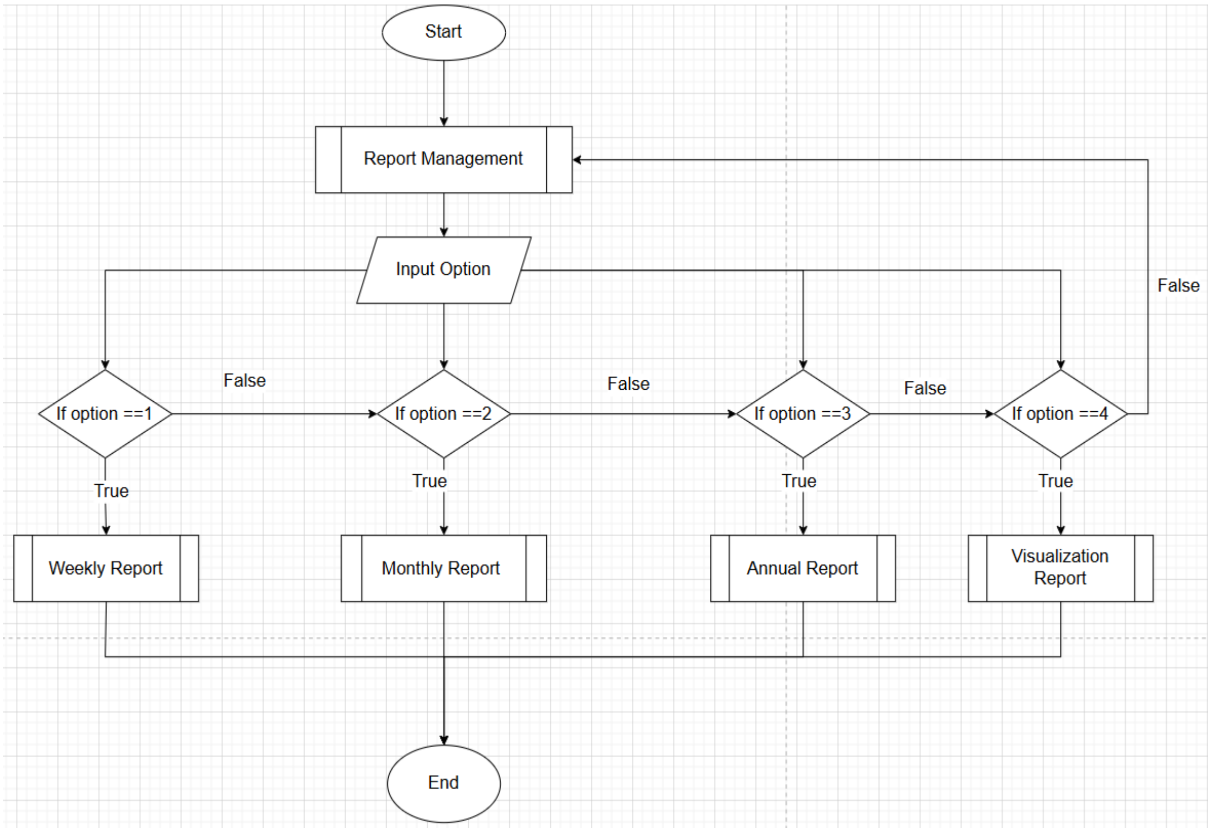

Figure 3.13 :Admin Report Management

User Management:

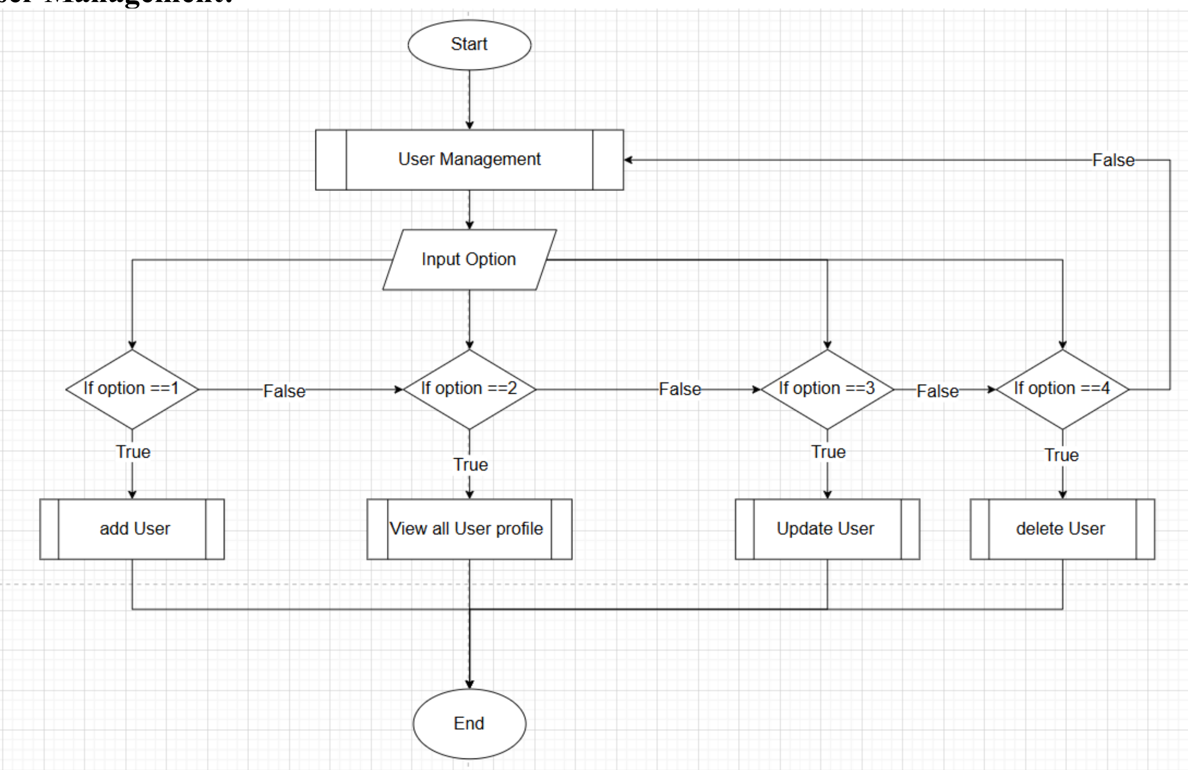

Figure 3.14 :Admin User Management

Admin add User:

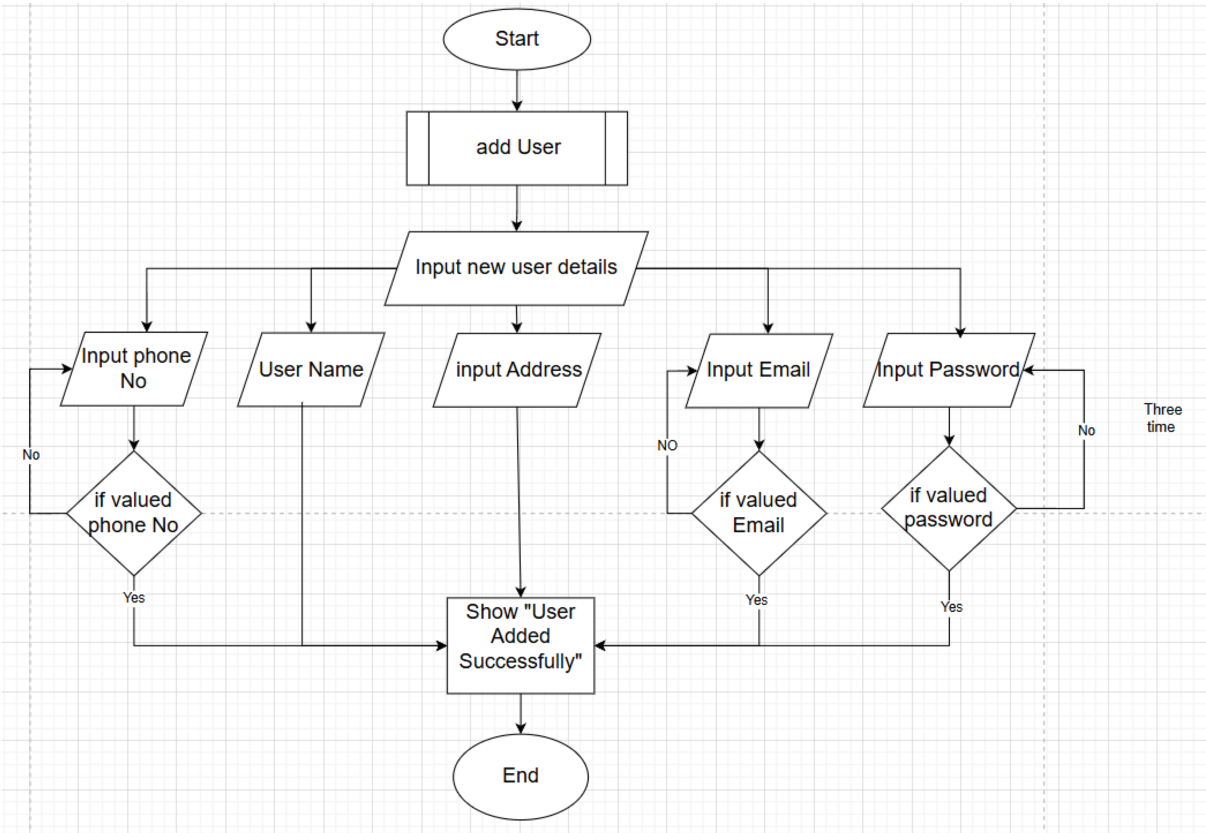

Figure 3.15 : Admin Add User

Admin view all users profile:

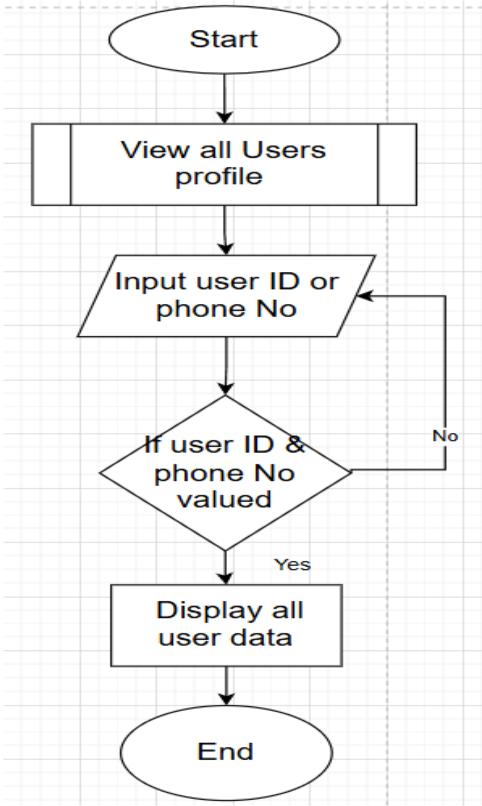

Figure 3.16 : Admin View User

Admin update user Info:

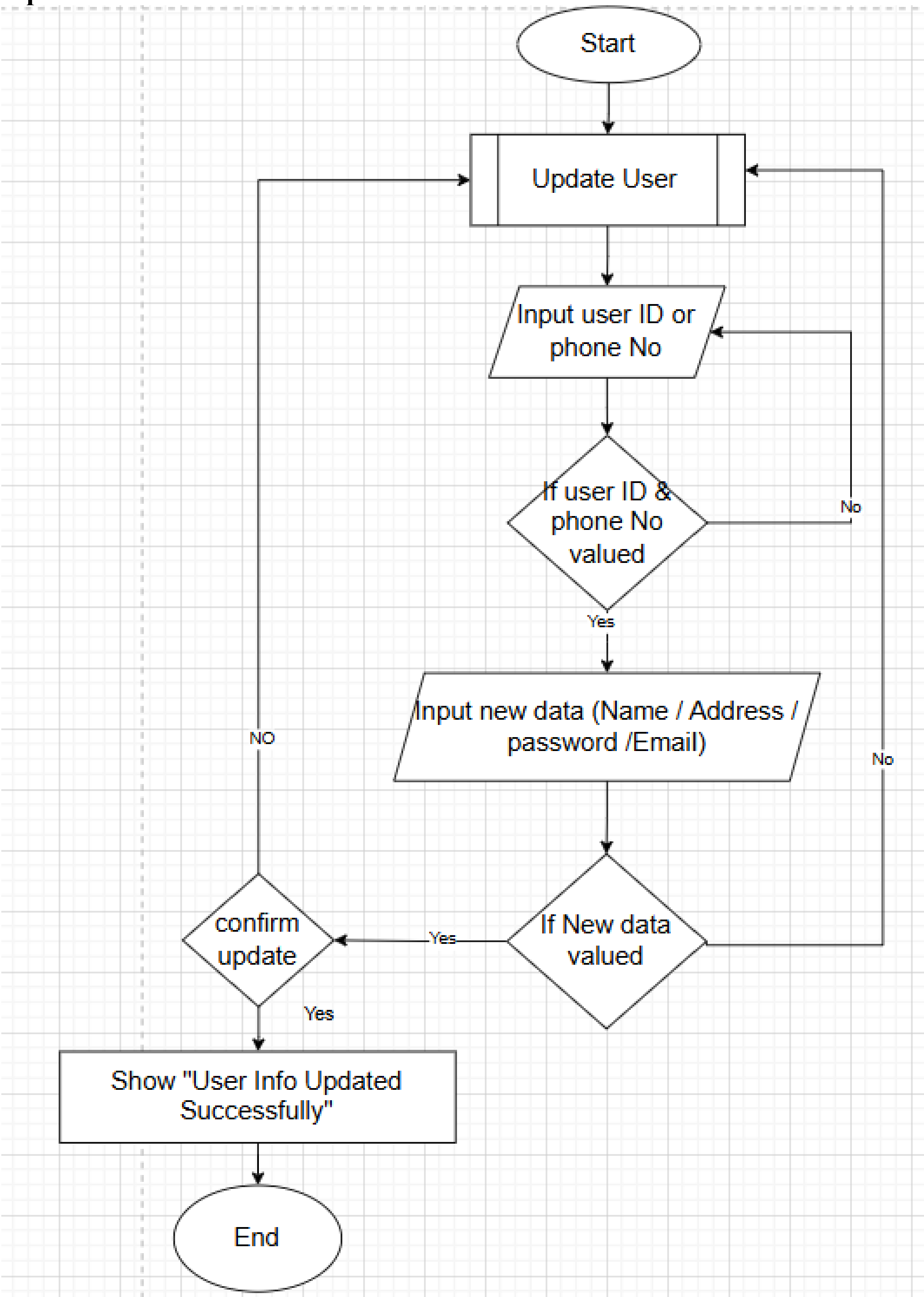

Figure 3.17 : Admin Update User Details

Admin delete user:

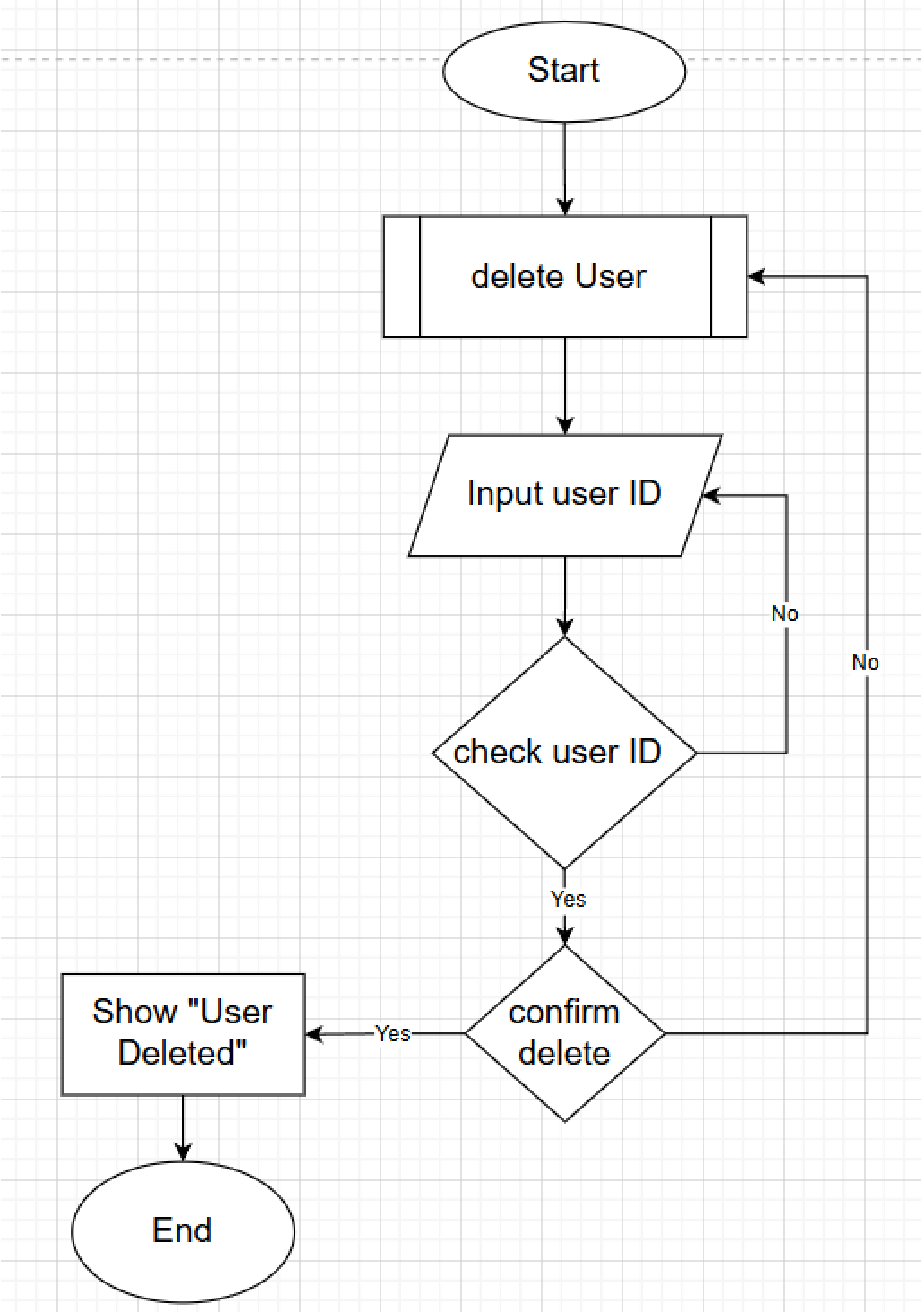

Figure 3.18 :Admin Delete User

User menu:

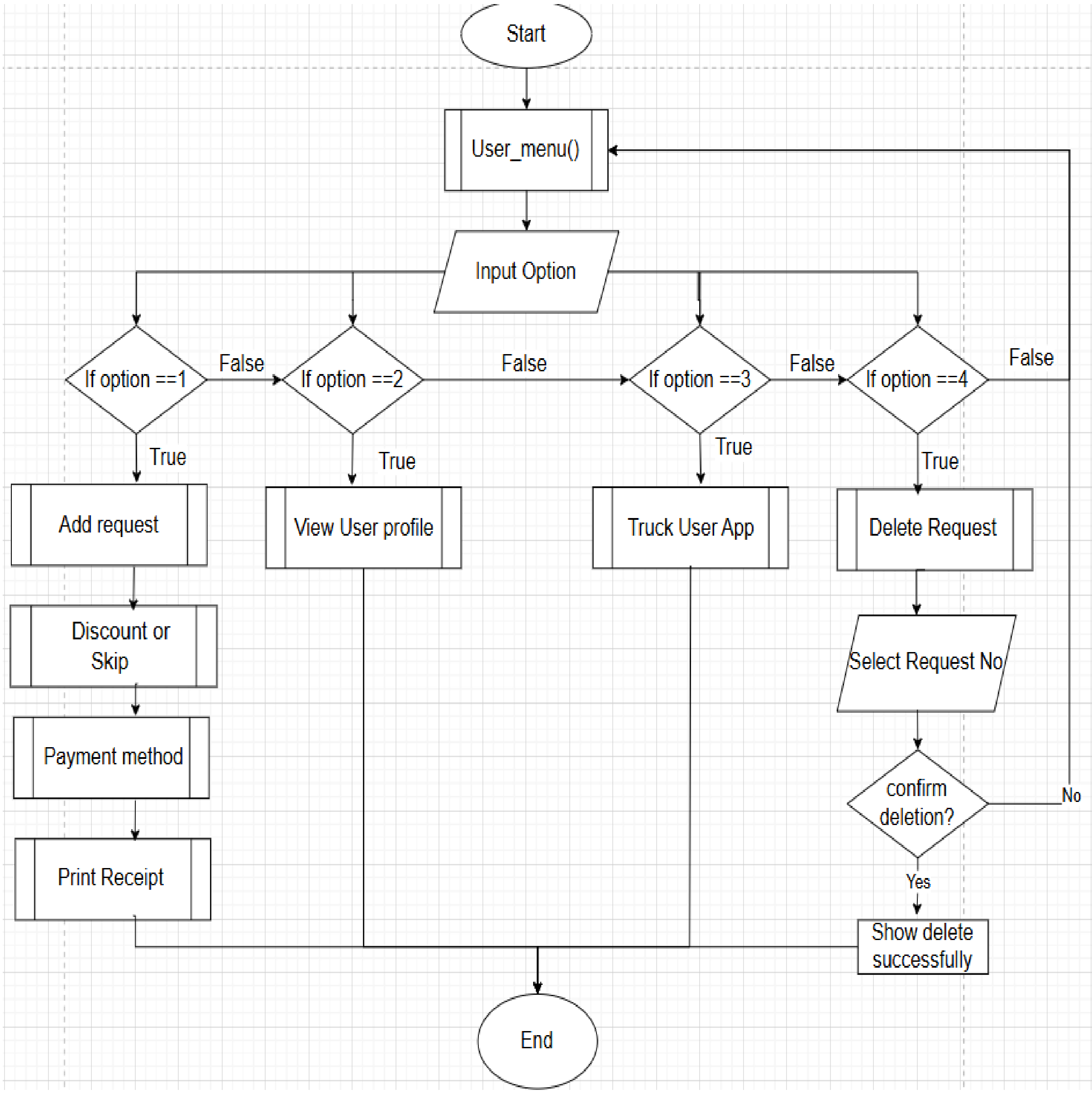

Figure 0.19 : Customer Menu

Add Request:

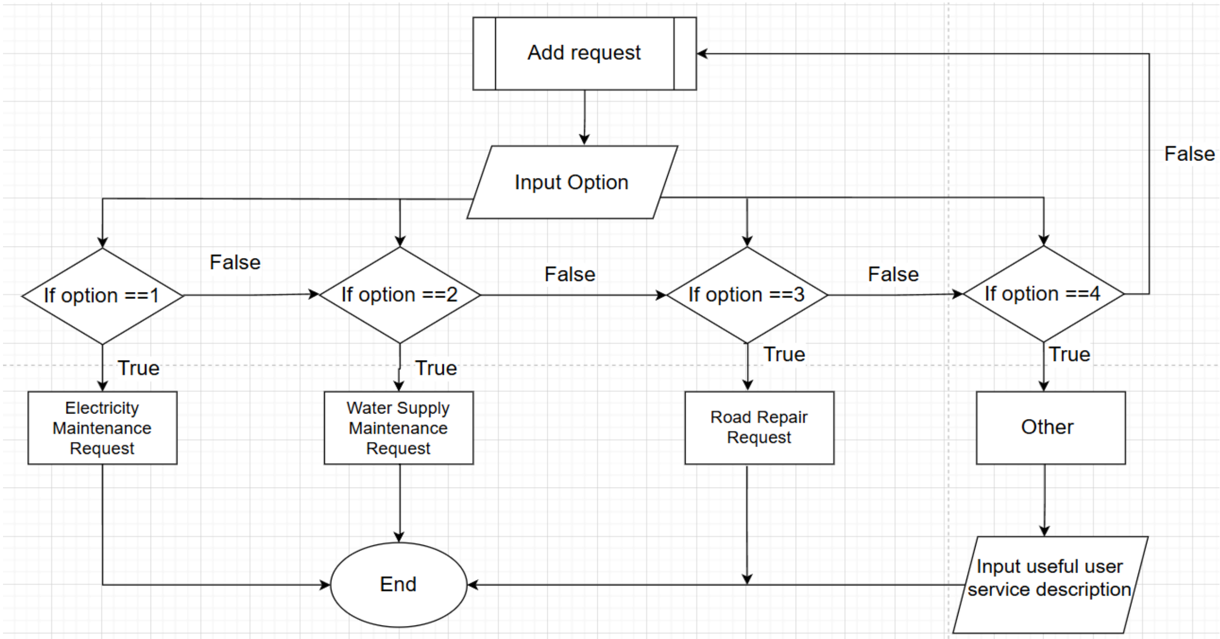

Figure 3.20 : Customer Add Request

Discount:

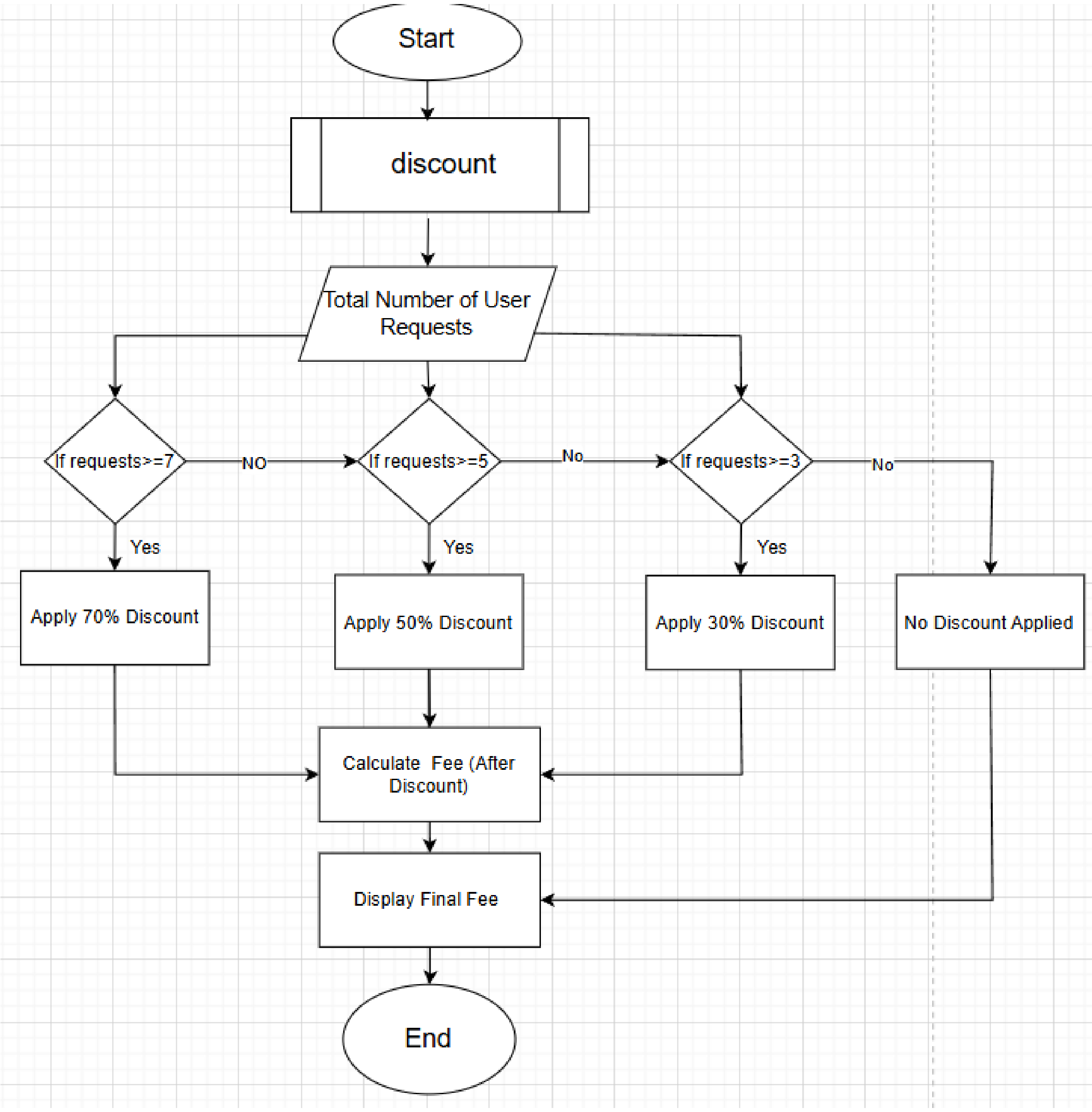

Figure 3.21 : Customer Discount

Payment method:

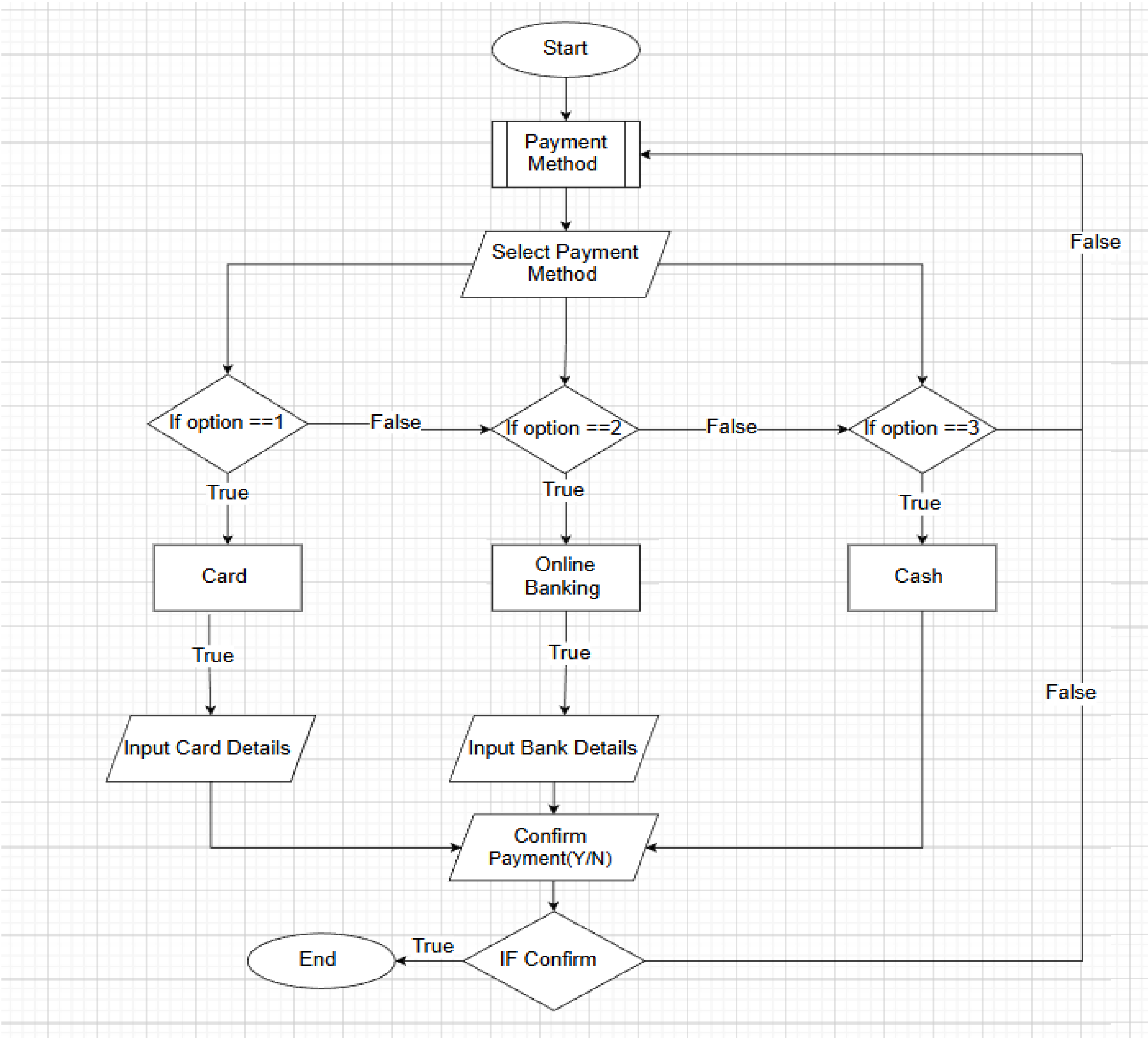

Figure 3.22 :Customer Payment Method

Print Receipt:

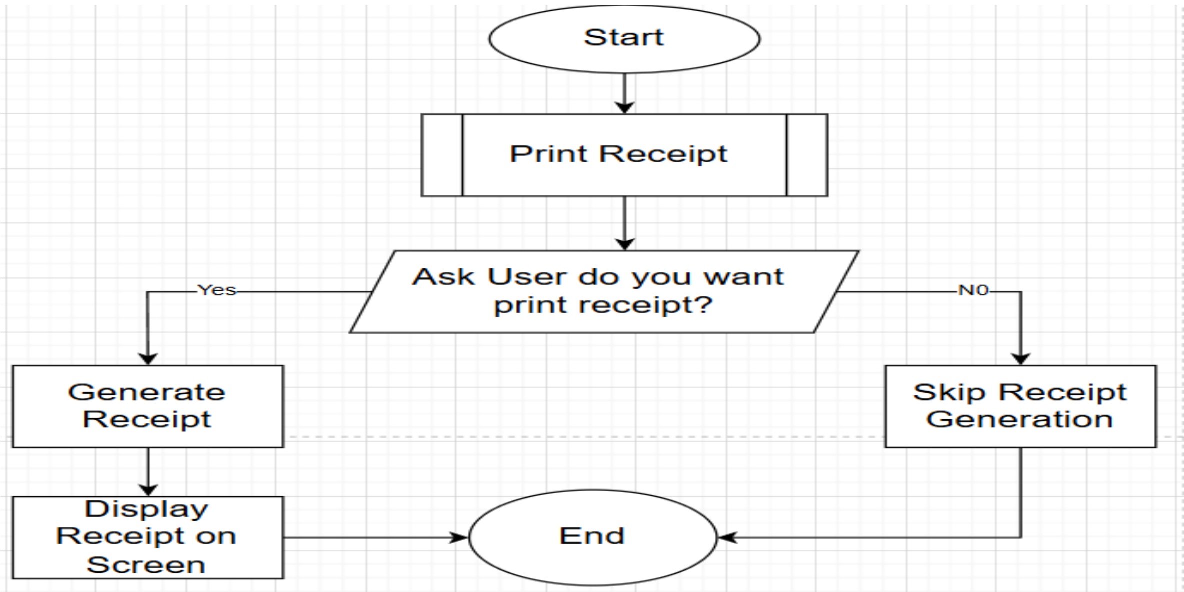

Figure 3.23 : Customer Receipt

User profile:

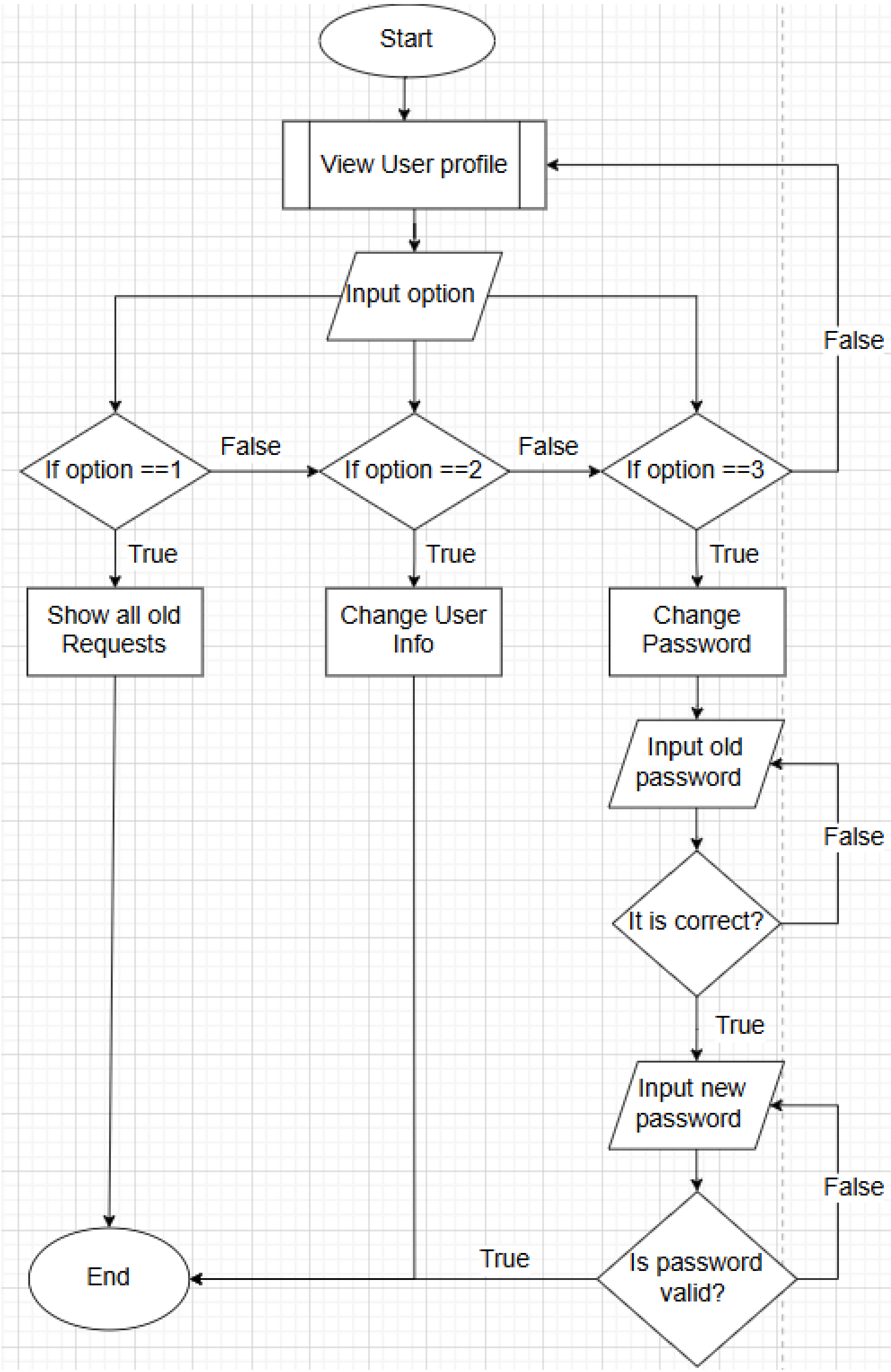

Figure 3.24 : Customer Profile

Track Customer App:

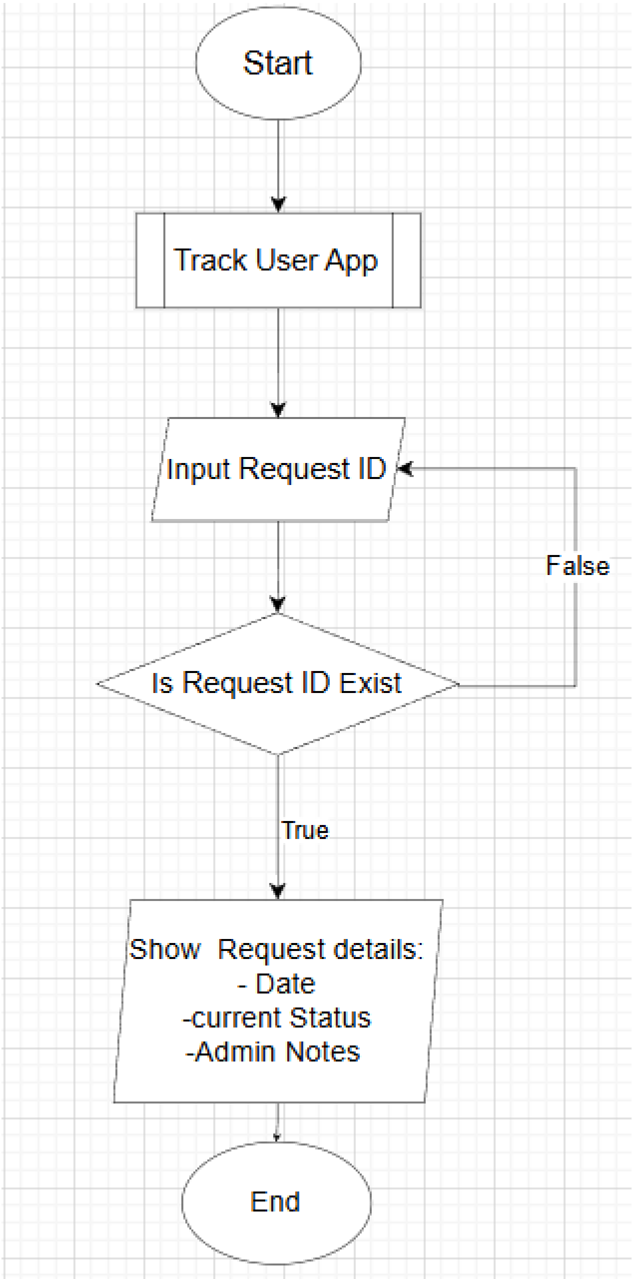

Figure 3.25 : Customer Track App

3.3 ERD

ERD:

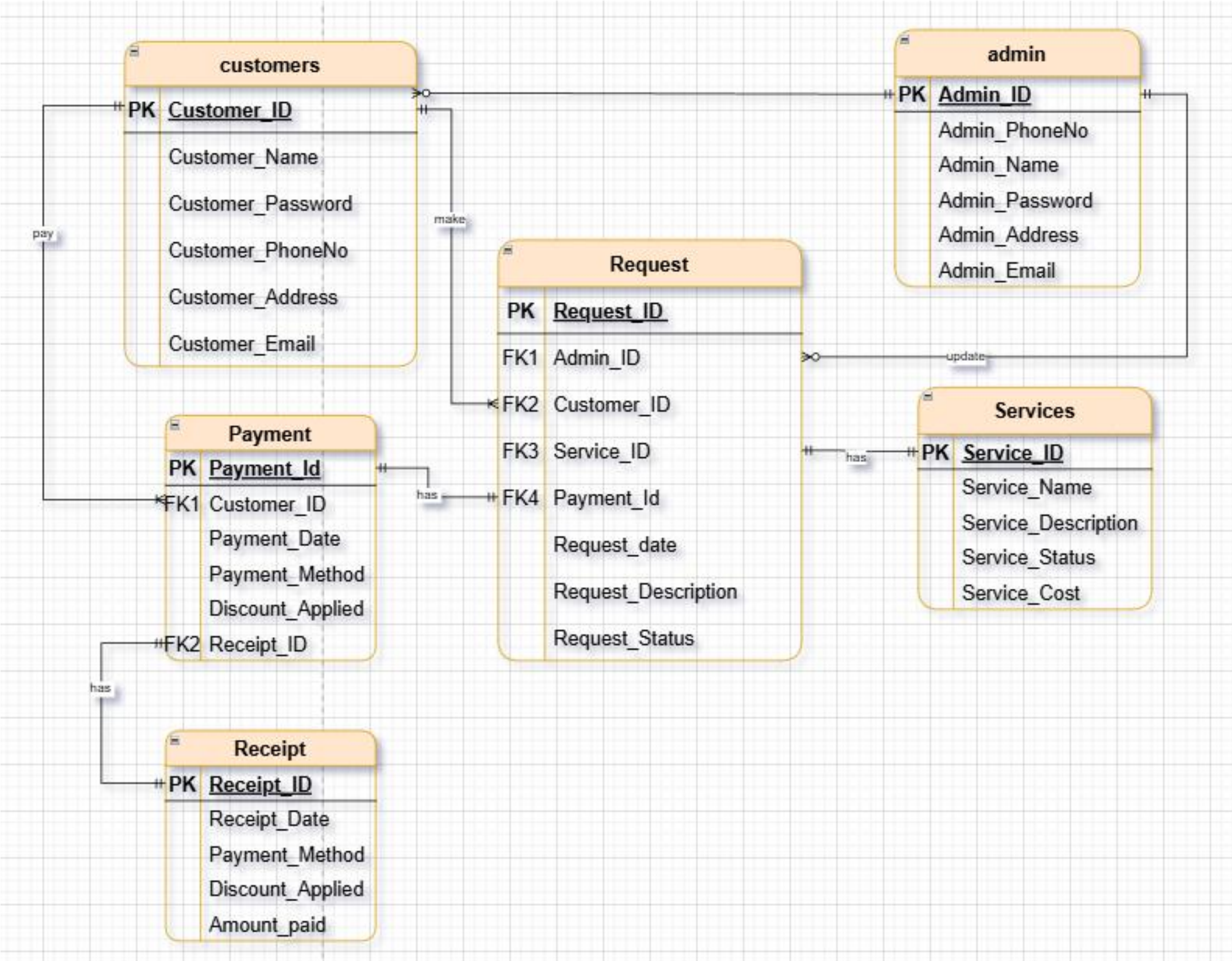

Figure 3.326: Entity Relationship Diagram

3.4 Data Normalization

The database design of the system has been normalized up to the Third Normal Form (3NF) to reduce data redundancy and improve data integrity.

First Normal Form (1NF) is achieved by ensuring that all attributes contain atomic values and that each table has a clearly defined primary key.

Second Normal Form (2NF) is satisfied by removing partial dependencies, where all non-key attributes depend fully on the primary key.

Third Normal Form (3NF) is achieved by eliminating transitive dependencies through separating related data into different tables such as User, Admin, Request, Service, Payment, Receipt, and Report.

This normalization process ensures efficient data storage, improves database consistency, and supports reliable and accurate system operations.

3.5 Data Dictionary

Admin:

Table 3.31: Admin Table

ATTRIBUTE NAME	CONTENT	DATA TYPE	UNIQUE	PRIMARY KEY (PK)/ FOREIGN KEY (FK)	FK REFERENCE TABLE
Admin_ID	System Admin Identification	VARCHAR (25)	YES	PK	-
Admin_Name	Admin Name	VARCHAR (45)	-	-	-
Admin_Email	Admin Email	VARCHAR (25)	YES	-	-
Admin_Password	Admin Password	VARCHAR (16)	-	-	-
Admin_Phone_No	Admin Phone Number	VARCHAR (25)	YES	-	-
Admin_Address	Admin Address	VARCHAR (255)	-	-	-

Customer:

Table 3.32: Customer Table

ATTRIBUTE NAME	CONTENT	DATA TYPE	UNIQUE	PRIMARY KEY (PK)/ FOREIGN KEY (FK)	FK REFERENCE TABLE
Customer_ID	Customer ID	VARCHAR (25)	YES	PK	-
Customer_Name	Customer Name	VARCHAR (45)	-	-	-
Customer_Email	Customer Email	VARCHAR (25)	YES	-	-
Customer_Password	Customer Password	VARCHAR (16)	-	-	-
Customer_Address	Customer Address	VARCHAR (255)	-	-	-
Customer_Phone_No	Customer Phone Number	VARCHAR (25)	YES	-	-

Service:

Table 3.33.3: Service Table

ATTRIBUTE NAME	CONTENT	DATA TYPE	UNIQUE	PRIMARY KEY (PK)/ FOREIGN KEY (FK)	FK REFERENCE TABLE
Service ID	Service_ID	VARCHAR (25)	YES	PK	-
Service_Name	Service Name	VARCHAR (45)	-	-	-
Service_Description	Service Description	VARCHAR (255)	-	-	-
Created_At	Created Date	DATE	-	-	-
Service_cost	Service cost	VARCHAR (25)	-	-	-
Service_Status	Service Status	VARCHAR(30)			

Request:

Table 3.33: Request Table

ATTRIBUTE NAME	CONTENET	Data type	UNIQUE	FK/PK	FK REFERENCE TABLE
Request_ID	Request ID	VARCHAR(25)	YES	PK	-
User_ID	Linked User	VARCHAR(25)	-	FK	User(User_ID)
Admin_ID	Assigned Admin	VARCHAR(25)	-	FK	Admin(Admin_ID)
Service_ID	Linked Service	VARCHAR(25)	-	FK	Service(Service_ID)
Request_Description	Description of Request	VARCHAR(255)	-	-	-
Request_Date	Date of Request	DATE	-	-	-
Request_Status	Status (Pending/In Progress/Completed)	VARCHAR(30)	-	-	-

Payment:

Table 3.34: Payment Table

ATTRIBUTE NAME	CONTENT	Data type	UNIQUE	FK/PK	FK REFERENCE TABLE
Payment_ID	Payment ID	VARCHAR(25)	YES	PK	-
User_ID	Linked User	VARCHAR(25)	-	FK	User(User_ID)
Request_ID	Linked Request	VARCHAR(25)	-	FK	Request(Request_ID)
Amount	Payment Amount	DECIMAL(10,2)	-	-	-
Discount_Applied	Discount Applied	DECIMAL(5,2)	-	-	-
Payment_Method	Payment Method	VARCHAR(30)	-	-	-
Payment_Date	Payment Date	DATE	-	-	-
Receipt_ID	Linked Receipt	VARCHAR(25)	-	FK	Receipt(Receipt_ID)

Receipt:

Table 3.35: Receipt Table

ATTRIBUTE NAME	CONTENT	Data type	UNIQUE	FK/PK	FK REFERENCE TABLE
Receipt_ID	Receipt ID	VARCHAR(25)	YES	PK	
Request_ID	Linked Request	VARCHAR(25)	-	FK	Request(Request_ID)
Amount_Paid	Amount Paid	DECIMAL(10,2)	-	-	
Payment_Method	Payment Method	VARCHAR(30)	-	-	
Receipt_Date	Date of Receipt	DATE	-	-	
Discount	Discount Percentage	DECIMAL(5,2)	-	-	

3.6 Interface Design

Main page:

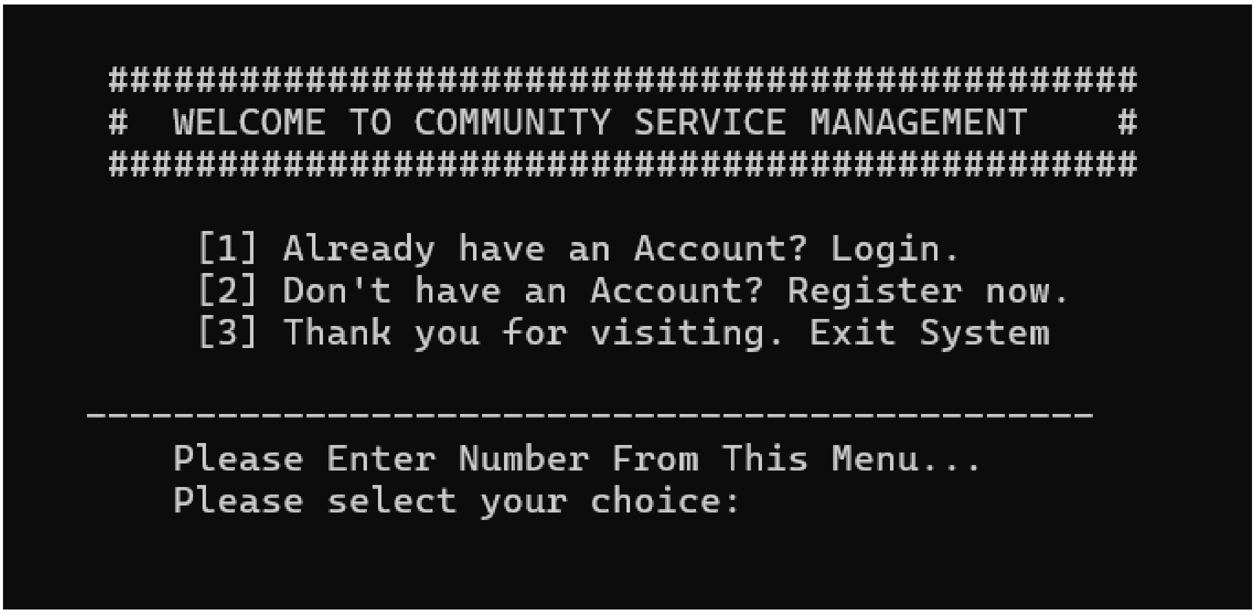

Figure 3.327: Main Page

Registration page:

```
=====================================================
#              ACCOUNT REGISTRATION                 #
=====================================================

        Select Account Type:
           [A] = Admin
           [C] = Customer
           [B] = Back to Main Menu

-----------------------------------------------------
Enter the letter corresponding to your account type.
Your Choice: a

Enter Admin Verification Code (Type bb to go back): A1234
Enter Full Name (Type bb to go back): Hussam Aldin Mahmoud
Enter Email (Type bb to go back): Hussam@gmail.com
Enter Phone Number (Type bb to go back): 0183660277
Enter Address (Type bb to go back): Melaka,Ayer keroh,Bukit Bruan

================ PASSWORD REQUIREMENTS ================
- Minimum 8 characters
- Maximum 16 characters
- Upper and lower letters
- At least 1 number
- Allowed special: @ # $ % & !
- (Type bb to go back)
=======================================================

Enter Password: Asdf@123
Confirm Password: Asdf@123

================ Confirm Your Information ================
Account ID : A36519
Name       : Hussam Aldin Mahmoud
Email      : Hussam@gmail.com
Phone      : 0183660277
Address    : Melaka,Ayer keroh,Bukit Bruan
Account    : Admin

Is this information correct? (Y/N): y

Registration Successful! Your ID is: A36519
```

Figure 3.328: Registration Page Module

Login page:

```
============================================================
#                    WELCOME TO LOGIN PAGE                 #
============================================================


        Select Account Type:
           [A] = Admin
           [C] = Customer
           [B] = Back to Main Menu

----------------------------------------------------
Enter the letter corresponding to your account type.
Your Choice: A
```

Figure 3.329 : Login Page module

```
============================================================
#                  WELCOME TO LOGIN PAGE                   #
============================================================


                Logging in as ADMIN
------------------------------------------------------------
        Enter Admin_ID(Type bb to go back): A16114
        Enter Password: ********
```

Figure 3.330 : Admin Login Page

Admin Menu:

```
Login Successful!
Welcome Admin: A16114!

============================================================
#                     ADMIN MAIN MENU                      #
============================================================
| 1. Request Management                                    |
------------------------------------------------------------
| 2. Service Management                                    |
------------------------------------------------------------
| 3. Customer Management                                   |
------------------------------------------------------------
| 4. Report Management                                     |
------------------------------------------------------------
| 5. My Profile                                            |
------------------------------------------------------------
| 6. Logout                                                |
------------------------------------------------------------

Please Enter Number From Given Options...
Select your choice:
```

Figure 3.331 : Admin Menu Module

Customer Login Page:

```
============================================================
#                 WELCOME TO LOGIN PAGE                    #
============================================================

                Logging in as CUSTOMER

------------------------------------------------------------
        Enter Customer_ID(Type bb to go back): c30460
        Enter Password: ********
```

Figure 3.32 : Customer Login Page

Customer Menu:

```
Login Successful!
Welcome Customer: c30460!

============================================================
#                        CUSTOMER MENU                     #
============================================================
| 1. Add Request                                           |
------------------------------------------------------------
| 2. My Profile                                            |
------------------------------------------------------------
| 3. Track Customer App                                    |
------------------------------------------------------------
| 4. Delete Request                                        |
------------------------------------------------------------
| 5. Logout                                                |
------------------------------------------------------------

Please select your choice:
```

Figure 3.33 : Customer Menu Module

Admin Request Management:

```
============================================================
                 REQUEST MANAGEMENT (ADMIN)
============================================================

 [1] View All Requests
------------------------------------------------------------
 [2] Update Request Status
------------------------------------------------------------
 [3] Search Request
------------------------------------------------------------
 [4] Delete Request
------------------------------------------------------------
 [5] Back
------------------------------------------------------------

Select your choice:
```

Figure 3.34 : Request Management Module

View All Request:

ALL REQUESTS

Request ID	Customer	Admin	Service	Description	Date	Status
R26324	c30460	A16114	S001	Electricity Maintenance	2026-01-11 07:57:58	In Progress
R29309	c30460	A16114	S002	Water Supply Maintenance	2026-01-11 08:13:03	Completed
R29374	c30460	A16114	S003	Road Repair	2026-01-11 08:14:52	Completed
R30037	c30461	A16114	S003	Road Repair	2026-01-11 08:16:43	Completed
R30517	c30461	A16114	S004	Other (Custom Service) - Sidewalk repair	2026-01-11 08:19:16	Completed
R30723	c30461	A16114	S002	Water Supply Maintenance	2025-11-11 08:20:07	Completed
R30778	c30461	A16114	S003	Road Repair	2025-11-11 08:20:25	Completed
R30827	c30461	A16114	S001	Electricity Maintenance	2025-12-11 08:21:01	Completed
R30938	c30461	A33701	S001	Electricity Maintenance	2025-12-11 08:21:39	Completed
R31196	c30461	A33701	S004	Other (Custom Service) - repair Benches	2025-12-11 08:22:41	Completed
R31552	c30462	A33701	S004	Other (Custom Service) - Traffic signal repair	2025-11-11 08:24:41	Completed
R31647	c30462	A33701	S001	Electricity Maintenance	2026-01-11 08:24:51	In Progress
R31676	c30462	A33701	S002	Water Supply Maintenance	2026-01-11 08:25:38	Pending
R31836	c30462	A33701	S003	Road Repair	2025-12-11 08:25:50	Completed
R39642	c30460	A16114	S001	Electricity Maintenance	2025-12-09 23:51:47	Completed

Figure 3.35 : View All Request Module

Update Request Status:

```
============================================ UPDATE REQUEST STATUS =================================================

+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| Request ID | Customer   | Admin      | Service      | Description                 | Date                | Status      |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R26324     | c30460     | A16114     | Electricity  | Electricity Maintenance     | 2026-01-11 07:57:58 | In Progress |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R29309     | c30460     | A16114     | Water Supply | Water Supply Maintenance    | 2026-01-11 08:13:03 | Completed   |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R29374     | c30460     | A16114     | Road Repair  | Road Repair                 | 2026-01-11 08:14:52 | Completed   |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30037     | c30461     | A16114     | Road Repair  | Road Repair                 | 2026-01-11 08:16:43 | Completed   |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30517     | c30461     | A16114     | Other        | Other (Custom Service) -    | 2026-01-11 08:19:16 | Completed   |
|            |            |            | (Custom      | Sidewalk repair             |                     |             |
|            |            |            | Service)     |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30723     | c30461     | A16114     | Water Supply | Water Supply Maintenance    | 2025-11-11 08:20:07 | Completed   |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30778     | c30461     | A16114     | Road Repair  | Road Repair                 | 2025-11-11 08:20:25 | Completed   |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30827     | c30461     | A16114     | Electricity  | Electricity Maintenance     | 2025-12-11 08:21:01 | Completed   |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R30938     | c30461     | A33701     | Electricity  | Electricity Maintenance     | 2025-12-11 08:21:39 | Completed   |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R31196     | c30461     | A33701     | Other        | Other (Custom Service) -    | 2025-12-11 08:22:41 | Completed   |
|            |            |            | (Custom      | repair Benches              |                     |             |
|            |            |            | Service)     |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R31552     | c30462     | A33701     | Other        | Other (Custom Service) -    | 2025-11-11 08:24:41 | Completed   |
|            |            |            | (Custom      | Traffic signal repair       |                     |             |
|            |            |            | Service)     |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R31647     | c30462     | A16114     | Electricity  | Electricity Maintenance     | 2026-01-11 08:24:51 | Pending     |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R31676     | c30462     | A33701     | Water Supply | Water Supply Maintenance    | 2026-01-11 08:25:38 | Pending     |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R31836     | c30462     | A33701     | Road Repair  | Road Repair                 | 2025-12-11 08:25:50 | Completed   |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
| R39642     | c30460     | A16114     | Electricity  | Electricity Maintenance     | 2025-12-09 23:51:47 | Completed   |
|            |            |            | Maintenance  |                             |                     |             |
+------------+------------+------------+--------------+-----------------------------+---------------------+-------------+
Enter Request ID (Type bb to go back): R31647

What do you want to do?
 1. Approve
 2. Reject
 0. Back
Your Choice: 1

Select Status after approval:
 1. Pending
 2. In Progress
 3. Completed
your Choice: 2

Apply changes? (Y/N): y

Status updated successfully!
Press Enter to return...View All Request:
```

Figure 3.36 : Update Request Status Module

Search Request:

```
================= SEARCH REQUEST ===============================

    +------------+-------------+---------------------+
    | Request ID | Status      | Date                |
    +------------+-------------+---------------------+
    | R26324     | In Progress | 2026-01-11 07:57:58 |
    | R29309     | Completed   | 2026-01-11 08:13:03 |
    | R29374     | Completed   | 2026-01-11 08:14:52 |
    | R30037     | Completed   | 2026-01-11 08:16:43 |
    | R30517     | Completed   | 2026-01-11 08:19:16 |
    | R30723     | Completed   | 2025-11-11 08:20:07 |
    | R30778     | Completed   | 2025-11-11 08:20:25 |
    | R30827     | Completed   | 2025-12-11 08:21:01 |
    | R30938     | Completed   | 2025-12-11 08:21:39 |
    | R31196     | Completed   | 2025-12-11 08:22:41 |
    | R31552     | Completed   | 2025-11-11 08:24:41 |
    | R31647     | In Progress | 2026-01-11 08:24:51 |
    | R31676     | Pending     | 2026-01-11 08:25:38 |
    | R31836     | Completed   | 2025-12-11 08:25:50 |
    | R39642     | Completed   | 2025-12-09 23:51:47 |
    +------------+-------------+---------------------+

    Enter Request ID (Type bb to go back): R26324
```

```
================== REQUEST DETAILS ==============================================

      +-----------------------+-----------------------------------------------+
      | Field                 | Value                                         |
      +-----------------------+-----------------------------------------------+
      | Request ID            | R26324                                        |
      | Customer              | c30460                                        |
      | Admin                 | A16114                                        |
      | Service               | Electricity Maintenance                       |
      | Status                | In Progress                                   |
      | Date                  | 2026-01-11 07:57:58                            |
      | Description           | Electricity Maintenance                      |
      +-----------------------+-----------------------------------------------+

      Press Enter to return...
```

Figure 3.37 : Search Request Module

Delete Request:

```
==================== DELETE REQUEST ==================================

        +-------------+-------------+---------------------+
        | Request ID | Status       | Date                |
        +-------------+-------------+---------------------+
        | R26324     | In Progress | 2026-01-11 07:57:58 |
        | R29309     | Completed   | 2026-01-11 08:13:03 |
        | R29374     | Completed   | 2026-01-11 08:14:52 |
        | R30037     | Completed   | 2026-01-11 08:16:43 |
        | R30517     | Completed   | 2026-01-11 08:19:16 |
        | R30723     | Completed   | 2025-11-11 08:20:07 |
        | R30778     | Completed   | 2025-11-11 08:20:25 |
        | R30827     | Completed   | 2025-12-11 08:21:01 |
        | R30938     | Completed   | 2025-12-11 08:21:39 |
        | R31196     | Completed   | 2025-12-11 08:22:41 |
        | R31552     | Completed   | 2025-11-11 08:24:41 |
        | R31647     | In Progress | 2026-01-11 08:24:51 |
        | R31676     | Pending     | 2026-01-11 08:25:38 |
        | R31836     | Completed   | 2025-12-11 08:25:50 |
        | R39642     | Completed   | 2025-12-09 23:51:47 |
        +-------------+-------------+---------------------+

        Enter Request ID to delete (Type bb to go back): R31676

        =================== REQUEST DETAILS ===================
         Request ID      : R31676
         Customer        : c30462
         Service         : Water Supply Maintenance
         Status          : Pending
         Date            : 2026-01-11 08:25:38
         Description     : Water Supply Maintenance
        =======================================================

        Are you sure you want to delete this request? (Y/N): y

        Request deleted successfully.
        Press Enter to go back...
```

Figure 3.38 : Delete Request Module

Admin Service Management:

```
=============================================================
                     SERVICE MANAGEMENT
=============================================================

 [1] View All Services
-------------------------------------------------------------
 [2] Add New Service
-------------------------------------------------------------
 [3] Update Existing Service
-------------------------------------------------------------
 [4] Delete Service
-------------------------------------------------------------
 [5] Search Service
-------------------------------------------------------------
 [6] Back
-------------------------------------------------------------

Select your choice:
```

Figure 3.39 : Service Management Module

View All Service:

```
                 ================================ ALL SERVICES ================================

+----------+-----------------------------+-------------------------------------------+-------+-----------+
|    ID    |            Name             |                Description                | Cost  |  Status   |
+----------+-----------------------------+-------------------------------------------+-------+-----------+
| S001     | Electricity Maintenance     | Fixing electrical issues and maintenance  | 40    | Active    |
| S002     | Water Supply Maintenance    | Repairing water pipeline and supply problems | 40 | Active    |
| S003     | Road Repair                 | Fixing roads and surface damages          | 60    | Active    |
| S004     | Other (Custom Service)      | Custom service depending on customer request | 70 | Active    |
+----------+-----------------------------+-------------------------------------------+-------+-----------+
```

Figure 3.340 : View All Service Module

Add Service:

```
=================================== ADD NEW SERVICE ===================================

        Generated Service ID: S005

        Enter Service Name (Type bb to cancel): Park Benches repair
        Enter Description (Type bb to cancel): Publik park Maintenance
        Enter Price (Numbers only) (Type bb to cancel): 35

        Service added successfully!
```

Figure 3.341 : Add Service Module

Update Service:

```
                    ================================= UPDATE SERVICE =================================

+----------+-----------------------------+--------------------------------------------+-------+-----------+
|    ID    |            Name             |                Description                 | Cost  |  Status   |
+----------+-----------------------------+--------------------------------------------+-------+-----------+
| S001     | Electricity Maintenance     | Fixing electrical issues and maintenance   | 40    | Active    |
| S002     | Water Supply Maintenance    | Repairing water pipeline and supply problems | 40  | Active    |
| S003     | Road Repair                 | Fixing roads and surface damages           | 60    | Active    |
| S004     | Other (Custom Service)      | Custom service depending on customer request | 70  | Active    |
| S005     | Park Benches repair         | Publik park Maintenance                    | 35    | Active    |
+----------+-----------------------------+--------------------------------------------+-------+-----------+

Enter Service ID to update (Type bb to cancel): S005

Current Data:
--------------------------------------------
Name       : Park Benches repair
Description: Publik park Maintenance
Cost       : RM 35
Status     : Active
--------------------------------------------

New Name (Type bb to cancel / leave empty to keep same):
New Description (Type bb to cancel / leave empty to keep same):
New Cost (Type bb to cancel  / leave empty to keep same): 45
New Status (Active / Inactive) (Type bb to cancel / leave empty to keep same): Inactive

Service updated successfully!
Press Enter to return...
```

Figure 3.342 : Update Service Module

Delete Service:

```
C:\Users\hussa\source\repos\

                    ================================= DELETE SERVICE =================================

+----------+-----------------------------+--------------------------------------------+-------+-----------+
|    ID    |            Name             |                Description                 | Cost  |  Status   |
+----------+-----------------------------+--------------------------------------------+-------+-----------+
| S001     | Electricity Maintenance     | Fixing electrical issues and maintenance   | 40    | Active    |
| S002     | Water Supply Maintenance    | Repairing water pipeline and supply problems | 40  | Active    |
| S003     | Road Repair                 | Fixing roads and surface damages           | 60    | Active    |
| S004     | Other (Custom Service)      | Custom service depending on customer request | 70  | Active    |
| S005     | Park Benches repair         | Publik park Maintenance                    | 45    | Inactive  |
+----------+-----------------------------+--------------------------------------------+-------+-----------+

Enter Service ID to delete (Type bb to cancel): S005

Are you sure you want to delete this service? (Y/N): y

Service deleted successfully!
Press Enter to continue...
```

Figure 3.343 : Delete Service Module

Search Service :

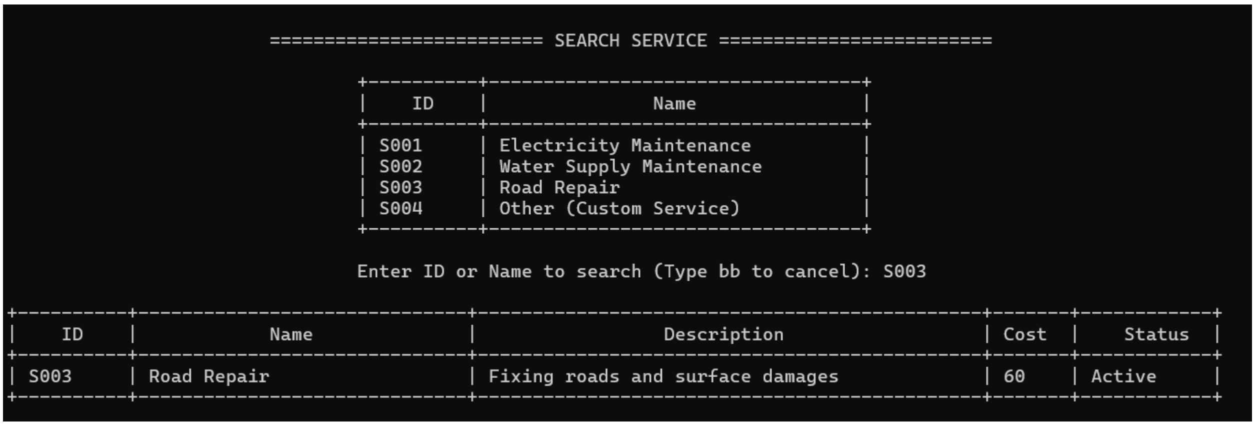

Figure 3.344 : Search Service Module

Admin Customer Management:

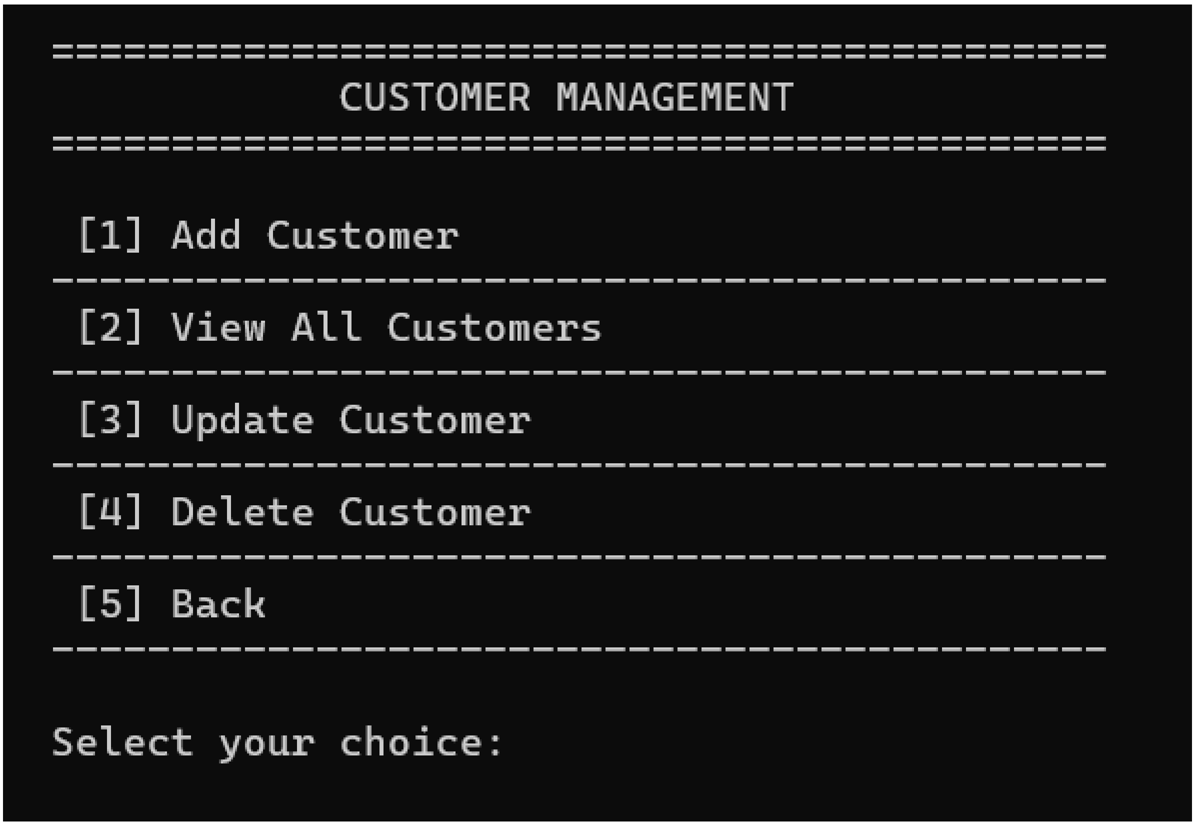

Figure 3.345 : Customer Management Module

View All Customer:

```
                    =================== ALL CUSTOMERS ===================

+----------+------------------------------+-------------+-----------------------------------------+----------------------------------------+
|   ID     |             Name             |    Phone    |                  Email                  |                Address                 |
+----------+------------------------------+-------------+-----------------------------------------+----------------------------------------+
| C30460   | Hussam Aldin Mahmoud         | 0183660298  | hus@gmai.com                            | Grand                                  |
| C30461   | Ahmad Khaled                 | 0183660301  | ahmad.khaled01@gmail.com                | Bukit Beruang, Melaka                  |
| C30462   | Mohammed Ali                 | 0183660302  | mohammed.ali02@gmail.com                | Ayer Keroh, Melaka                     |
| C30463   | Omar Hassan                  | 0183660303  | omar.hassan03@gmail.com                 | Batu Berendam, Melaka                  |
| C30464   | Youssef Ahmad                | 0183660304  | youssef.ahmad04@gmail.com               | Cheng, Melaka                          |
| C30465   | Khaled Mahmoud               | 0183660305  | khaled.mahmoud05@gmail.com              | Klebang, Melaka                        |
| C30466   | Ali Nasser                   | 0183660306  | ali.nasser06@gmail.com                  | Taman Merdeka, Melaka                  |
| C30467   | Hassan Ibrahim               | 0183660307  | hassan.ibrahim07@gmail.com              | Ujong Pasir, Melaka                    |
| C30468   | Mustafa Saleh                | 0183660308  | mustafa.saleh08@gmail.com               | Tanjung Kling, Melaka                  |
| C30469   | Fadi Rahman                  | 0183660309  | fadi.rahman09@gmail.com                 | Durian Tunggal, Melaka                 |
| C30470   | Samer Tarek                  | 0183660310  | samer.tarek10@gmail.com                 | Masjid Tanah, Melaka                   |
+----------+------------------------------+-------------+-----------------------------------------+----------------------------------------+

Press Enter to return...
```

Figure 3.346 : View All Customer Module

Admin Add Customer :

```
===================== ADD CUSTOMER =====================

Enter Customer Name (Type bb to go back): Ahmad Ali
Enter Email (Type bb to go back): Ahmad@gmail.com
Enter Phone Number (Type bb to go back): 0183660222
Enter Address (Type bb to go back): Melaka,Ayer keroh,Jalan M9
================ PASSWORD REQUIREMENTS ================
- Minimum 8 characters
- Maximum 16 characters
- Upper and lower letters
- At least 1 number
- Allowed special: @ # $ % & !
- (Type bb to go back)
=======================================================

Enter Password: Asdf@123
Confirm Password: Asdf@123

Customer Added Successfully!
Generated ID: C3047

Press Enter to return...
```

Figure 3.347 : Admin Add Customer Module

Admin Update Customer:

```
                                   ================== UPDATE CUSTOMER ==================

+----------+-----------------------+--------------+----------------------------------+-----------------------+-------------------+
|    ID    |         Name          |    Phone     |              Email               |        Address        |     Password      |
+----------+-----------------------+--------------+----------------------------------+-----------------------+-------------------+
| C30460   | Hussam Aldin Mahmoud | 0183660298   | hus@gmai.com                     | Grand                 | Asdf@123          |
| C30461   | Ahmad Khaled          | 0183660301   | ahmad.khaled01@gmail.com         | Bukit Beruang, Me...  | Asdf@123          |
| C30462   | Mohammed Ali          | 0183660302   | mohammed.ali02@gmail.com         | Ayer Keroh, Melaka    | Asdf@123          |
| C30463   | Omar Hassan           | 0183660303   | omar.hassan03@gmail.com          | Batu Berendam, Me...  | Asdf@123          |
| C30464   | Youssef Ahmad         | 0183660304   | youssef.ahmad04@gmail.com        | Cheng, Melaka         | Asdf@123          |
| C30465   | Khaled Mahmoud        | 0183660305   | khaled.mahmoud05@gmail.com       | Klebang, Melaka       | Asdf@123          |
| C30466   | Ali Nasser            | 0183660306   | ali.nasser06@gmail.com           | Taman Merdeka, Me...  | Asdf@123          |
| C30467   | Hassan Ibrahim        | 0183660307   | hassan.ibrahim07@gmail.com       | Ujong Pasir, Melaka   | Asdf@123          |
| C30468   | Mustafa Saleh         | 0183660308   | mustafa.saleh08@gmail.com        | Tanjung Kling, Me...  | Asdf@123          |
| C30469   | Fadi Rahman           | 0183660309   | fadi.rahman09@gmail.com          | Durian Tunggal, M...  | Asdf@123          |
| C3047    | Ahmad Ali             | 0183660222   | Ahmad@gmail.com                  | Melaka,Ayer keroh...  | Asdf@123          |
| C30470   | Samer Tarek           | 0183660310   | samer.tarek10@gmail.com          | Masjid Tanah, Melaka  | Asdf@123          |
+----------+-----------------------+--------------+----------------------------------+-----------------------+-------------------+

Enter Customer ID to update (Type bb to cancel): C3047

What do you want to update?
 1. Phone
 2. Email
 3. Address
 4. Password
 5. Back
Select your choice: 1
Enter new phone (Type bb to cancel): 0183660111

Updated successfully!
You can continue updating other details,
or choose (6) Back when you have finished.


What do you want to update?
 1. Phone
 2. Email
 3. Address
 4. Password
 5. Back
Select your choice:
```

Figure 3.348 : Admin Update Customer Module

Admin Delete Customer:

```
        ================= DELETE CUSTOMER ==================

+----------+----------------------+-------------+--------------------------------+
|    ID    |         Name         |    Phone    |            Email               |
+----------+----------------------+-------------+--------------------------------+
| C30460   | Hussam Aldin Mahmoud | 0183660298  | hus@gmai.com                   |
| C30461   | Ahmad Khaled         | 0183660301  | ahmad.khaled01@gmail.com       |
| C30462   | Mohammed Ali         | 0183660302  | mohammed.ali02@gmail.com       |
| C30463   | Omar Hassan          | 0183660303  | omar.hassan03@gmail.com        |
| C30464   | Youssef Ahmad        | 0183660304  | youssef.ahmad04@gmail.com      |
| C30465   | Khaled Mahmoud       | 0183660305  | khaled.mahmoud05@gmail.com     |
| C30466   | Ali Nasser           | 0183660306  | ali.nasser06@gmail.com         |
| C30467   | Hassan Ibrahim       | 0183660307  | hassan.ibrahim07@gmail.com     |
| C30468   | Mustafa Saleh        | 0183660308  | mustafa.saleh08@gmail.com      |
| C30469   | Fadi Rahman          | 0183660309  | fadi.rahman09@gmail.com        |
| C3047    | Ahmad Ali            | 0183660111  | Ahmad@gmail.com                |
| C30470   | Samer Tarek          | 0183660310  | samer.tarek10@gmail.com        |
+----------+----------------------+-------------+--------------------------------+

Enter Customer ID to delete (bb to cancel): C3047

-----------------------------------------
Customer Name : Ahmad Ali
Phone         : 0183660111
Email         : Ahmad@gmail.com
-----------------------------------------

Are you sure you want to delete this customer? (Y/N): y

Customer deleted successfully!
Press Enter...
```

Figure 3.349 : Admin Delete Customer Module

Admin Report Management:

```
==================================================
               REPORT MANAGEMENT
==================================================

 [1] Weekly Report
--------------------------------------------------
 [2] Monthly Report
--------------------------------------------------
 [3] Annual Report
--------------------------------------------------
 [4] Data Visualization
--------------------------------------------------
 [5] Back
--------------------------------------------------

Please select your choice:
```

Figure 3.350 : Admin Report Management Module

Weekly Report:

```
========================================
              Weekly Report
========================================

           ===Service Report===
+-----------+---------------+----------------+
| Service ID| Num Requests  | Total Cost(RM) |
+-----------+---------------+----------------+
| S002      | 1             | 40.00          |
| S003      | 2             | 120.00         |
| S004      | 1             | 70.00          |
+-----------+---------------+----------------+

Average Service Cost: 76.67 RM
Minimum Service Cost: 40.00 RM
Maximum Service Cost: 120.00 RM

Press Enter to go back...
```

Figure 3.351 : Weekly Report Module

Monthly Report:

```
                ===Monthly Report===

-----------------------------------------------
Enter month (1-12 or Name)(Type bb for back): 12
Enter year(Type bb for back): 2025
```

```
========================================
            Monthly Report
========================================

            Report for: 12/2025

         ===Service Report===
+-----------+---------------+----------------+
| Service ID| Num Requests  | Total Cost(RM) |
+-----------+---------------+----------------+
| S001      | 3             | 120.00         |
| S003      | 1             | 60.00          |
| S004      | 1             | 70.00          |
+-----------+---------------+----------------+

Average Service Cost: 83.33 RM
Minimum Service Cost: 60.00 RM
Maximum Service Cost: 120.00 RM

Press Enter to go back...
```

Figure 3.352 : Monthly Report Module

Annual Report:

```
===Annual Report==

----------------------------------
Enter year(Type bb for back): 2025
```

```
========================================
              Annual Report
========================================

           ===Service Report===
+-----------+---------------+----------------+
| Service ID| Num Requests  | Total Cost(RM) |
+-----------+---------------+----------------+
| S001      | 3             | 120.00         |
| S002      | 1             | 40.00          |
| S003      | 2             | 120.00         |
| S004      | 2             | 140.00         |
+-----------+---------------+----------------+

Average Service Cost: 105.00 RM
Minimum Service Cost: 40.00 RM
Maximum Service Cost: 140.00 RM

Press Enter to go back...
```

Figure 3.353 : Annual Report Module

Data Visualization:

```
================ DATA VISUALIZATION ================


Request Status Overview (Graphical)
-----------------------------------------
Pending   : ** (2)
Completed : ************** (14)
Rejected  : ** (2)

Total Requests : 18

Press Enter to return...
```

Figure 3.354 : Data Visualization Module

Admin My Profile:

```
============================================================
                         MY PROFILE
============================================================

 [1] View Profile
------------------------------------------------------------
 [2] Update Phone
------------------------------------------------------------
 [3] Update Address
------------------------------------------------------------
 [4] Update Password
------------------------------------------------------------
 [5] Update Email
------------------------------------------------------------
 [6] Back
------------------------------------------------------------

Select your choice:
```

Figure 3.355 : Admin My Profile Module

View My Profile:

```
 ==================== YOUR PROFILE ====================

 Admin ID    : A16114
 Name        : Hussam Aldin Mahmoud
 Phone       : 0183660281
 Address     : Ayer Keroh,Bukit Bruan
 Email       : hussammmahmoud@gmail.com

 Press Enter to return...
```

Figure 3.356 : View My Profile Module

Admin Up date Phone Number:

```
 ====================UPDATE PHONE ====================

 Current phone : 0183660281

 Enter new phone (10 to 12 digits) (Type bb to back): 0183660333

 Phone updated successfully!
 Press Enter to return...
```

Figure 3.357 : Up date Phone Number Module

Customer Add Request:

```
====================== ADD NEW REQUEST =====================

Available Services:
------------------------------------------------------------
[1] Electricity Maintenance  (RM 40.00)
[2] Water Supply Maintenance  (RM 40.00)
[3] Road Repair  (RM 60.00)
[4] Other (Custom Service)  (RM 70.00)
------------------------------------------------------------
Select your service (1-4) or (Type bb to go back): 1

Payment Method:
1. Online Banking
2. Card
3. Cash
Select your choice(Type bb to go back): 1
Enter Bank Name (Type bb to go back): Bank Islam
Enter Account ID (Type bb to go back): 12345678

Confirm this request? (Y/N): y

Request added successfully!!
---------------------------------

===================== RECEIPT =====================

Receipt ID       : RC28802
Request ID       : R21042
Customer ID      : c30460
Service Type     : Electricity Maintenance
Base Price       : RM 40.00
Discount Applied : RM 20.00
Final Price      : RM 20.00
Payment Method   : Online Banking

===================================================

Your request is being processed...
Please wait while our team reviews it.

Press Enter to return...
```

Figure 3.358 : Customer Add Request Module

Customer My Profile:

```
=============================================================
                         MY PROFILE
=============================================================

 [1] View Profile
-------------------------------------------------------------
 [2] Update Phone
-------------------------------------------------------------
 [3] Update Address
-------------------------------------------------------------
 [4] Update Password
-------------------------------------------------------------
 [5] Update Email
-------------------------------------------------------------
 [6] Back
-------------------------------------------------------------

Select your choice:
```

Figure 3.359 : Customer My Profile Module

Customer View Profile:

```
============================================
#             CUSTOMER PROFILE             #
============================================
Customer ID   : c30460
Name          : Hussam Aldin Mahmoud
Phone         : 0183660298
Email         : hus@gmai.com
Address       : Grand

----------- Request History -----------

Request ID : R21042
Service    : Electricity Maintenance
Description: Electricity Maintenance
Status     : Pending
Date       : 2026-01-11 10:18:25
--------------------------------------

Request ID : R29374
Service    : Road Repair
Description: Road Repair
Status     : Completed
Date       : 2026-01-11 08:14:52
--------------------------------------

Request ID : R29309
Service    : Water Supply Maintenance
Description: Water Supply Maintenance
Status     : Completed
Date       : 2026-01-11 08:13:03
--------------------------------------

Request ID : R26324
Service    : Electricity Maintenance
Description: Electricity Maintenance
Status     : In Progress
Date       : 2026-01-11 07:57:58
--------------------------------------

Request ID : R39642
Service    : Electricity Maintenance
Description: Electricity Maintenance
Status     : Completed
Date       : 2025-12-09 23:51:47
--------------------------------------

Press Enter to return...
```

Figure 3.360 : Customer View Profile Module

Customer Up date Address:

```
============================================
#           UpDate Address                #
============================================
Current: Grand

Enter new address (Type bb to go back): Melaka,Ayer keroh,Bucit Katile

Address updated successfully!
Press Enter to return...
```

Figure 3.361 : Customer Update Address Module

Customer Track App:

```
==================================================
#             Track Your Service Request         #
==================================================
Please enter your Request ID to view status
(Ex: R12345).

(Type bb to go back)


Your requests:

+------------+--------------+----------------------+
| RequestID  | Status       | Date                 |
+------------+--------------+----------------------+
| R21042     | Pending      | 2026-01-11 10:18:25  |
| R26324     | In Progress  | 2026-01-11 07:57:58  |
| R29309     | Completed    | 2026-01-11 08:13:03  |
| R29374     | Completed    | 2026-01-11 08:14:52  |
| R39642     | Completed    | 2025-12-09 23:51:47  |
+------------+--------------+----------------------+

Enter Request ID: R21042

----------- REQUEST DETAILS -----------
Request ID    : R21042
Service       : Electricity Maintenance
Description   : Electricity Maintenance
Status        : Pending
Request Date  : 2026-01-11 10:18:25
Paid Amount   : RM 20.00
Discount      : RM 20.00
--------------------------------------

Press Enter to continue...
```

Figure 3.362 : Customer Track App Module

Customer Delete Request:

```
===============================================
#              DELETE SERVICE REQUEST          #
===============================================

Your requests:

-----------------------------------------------
|  ID        |   Status        |    Date       |
-----------------------------------------------
|  R21042  |  Pending  |  2026-01-11 10:18:25 |
|  R26324  |  In Progress  |  2026-01-11 07:57:58 |
|  R29309  |  Completed  |  2026-01-11 08:13:03 |
|  R29374  |  Completed  |  2026-01-11 08:14:52 |
|  R39642  |  Completed  |  2025-12-09 23:51:47 |
-----------------------------------------------

Enter Request ID to delete (Type bb to go back): R21042

Are you sure you want to delete? (Y/N): Y

Request deleted successfully.
Press Enter to return...
```

Figure 3.363 : Customer Delete Request Module

CHAPTER 4: IMPLEMENTATION

4.1 Naming Convention

A set of rules and standards is used in this system to name variables, functions, and database elements consistently. Following proper naming conventions helps improve code readability, maintainability, and overall understanding of the system.

In this project, meaningful and descriptive names are used to clearly represent the purpose of each function and variable. Function and variable names follow the Camel Case naming style, while case sensitivity is maintained throughout the system to avoid ambiguity.

For database design, underscores (_) are used to separate words in table and attribute names, such as Customer_ID, Request_Status, and Service_ID. This approach improves clarity and consistency in database structure.

Overall, the adopted naming convention ensures that the system is well-organized, easy to read, and simple to maintain. Examples of the applied naming convention can be seen in the implemented functions as illustrated in the related figures**.**

```
void Main_Menu();
void registerUser();
void login_page();
void login_page(string accountType);
string HiddenPassword();
void admin_menu(string adminID);
void admin_requestManagement(string adminID);
void viewAllRequests();
void updateRequestStatus(string adminID);
void searchRequest();
void deleteRequest();
void admin_serviceManagement(string adminID);
void viewAllServices();
void addNewService();
void updateService();
void deleteService();
void searchService();
void admin_customerManagement(string adminID);
void addCustomer();
void viewAllCustomers();
void updateCustomer();
void deleteCustomer();
void admin_reportManagement(string adminID);
void serviceSalesReport(const string& title,const string& dateCondition,const string& subtitle);
void dataVisualization();
void annualReport();
void monthlyReport();
void weeklyReport();
void admin_profile(string adminID);
void viewAdminProfile(string adminID);
void updateAdminPhone(string adminID);
void updateAdminAddress(string adminID);
void updateAdminPassword(string adminID);
void updateAdminEmail(string adminID);
```

```cpp
void updateAdminEmail(string adminID);
void customer_menu(string customerID);
void addRequest(string customerID);
void customerProfileMenu(string customerID);
void viewCustomerProfile(string customerID);
void updateCustomerEmail(string customerID);
void updateCustomerPassword(string customerID);
void updateCustomerAddress(string customerID);
void updateCustomerPhone(string customerID);
void trackCustomerRequest(string customerID);
void deleteRequest(string customerID);
bool doesValueExist(const string& column, const string& value, const string& tableName);
bool isEmailExistsGlobal(const string& email);
bool isPhoneExistsGlobal(const string& phone);
bool isEmpty(const string& s);
bool isOnlyDigits(const string& s);
bool isOnlyLetters(const string& s);
string trim(const string& s);
```

Figure 4.1:Function

4.2 Function

One of the examples of function checks the existence of a value in a database table. It is mainly used for validation purposes to avoid duplicate records and ensure data consistency in the system.

```cpp
// Function to check if a value exists in a table
bool doesValueExist(const string& column, const string& value, const string& tableName) {
    try {
        sql::Statement* stmt = globalCon->createStatement();

        string query = "SELECT " + column +
            " FROM " + tableName +
            " WHERE " + column + " = '" + value + "' LIMIT 1";

        sql::ResultSet* res = stmt->executeQuery(query);

        bool exists = res->next(); // True if found

        delete res;
        delete stmt;

        return exists;
    }
    catch (sql::SQLException& e) {
        cout << "SQL Error: " << e.what() << endl;
        return false;
    }
}
```

Figure 4.2: Function Example

4.3 Vector

Arrays are used to store and manage collections of data. In this system, vectors are used to store service information such as IDs, names, and prices for display and selection purposes. Example from the code.

```
cout << "\n";
cout << "\t\t\t\t====================== ADD NEW REQUEST ====================\n\n";

vector<string> ids, names;
vector<double> prices;

string q = "SELECT Service_ID, Service_Name, Service_Cost FROM service WHERE Status='Active'";
sql::Statement* stmt = globalCon->createStatement();
sql::ResultSet* res = stmt->executeQuery(q);

cout << "\t\t\t\tAvailable Services:\n";
cout << "\t\t\t\t------------------------------------------------------------\n";

while (res->next())
{
    ids.push_back(res->getString("Service_ID"));
    names.push_back(res->getString("Service_Name"));
    prices.push_back(res->getDouble("Service_Cost"));
}

delete res;
delete stmt;
```

Figure 04.3: Vector

4.4 Selection

Selection is used in the system to control the program flow based on user input. The following code shows an example of selection using if and else if statements, where the system checks the selected menu option and executes the corresponding admin function.

If an invalid option is entered, an error message is displayed and the user is asked to try again.

```
if (choice == "1") {
    system("cls");
    admin_requestManagement(adminID);
}
else if (choice == "2") {
    system("cls");
   admin_serviceManagement(adminID);
}
else if (choice == "3") {
    system("cls");
    admin_customerManagement(adminID);
}
else if (choice == "4") {
    system("cls");
    admin_reportManagement(adminID);
}
else if (choice == "5") {
    system("cls");
    admin_profile(adminID);
}
else if (choice == "6") {
    cout << "\n";
    cout << "\t\t\t\tLogging out...\n";
    break;
}
else {
    cout << "\n";
    cout << "\t\t\t\tInvalid choice! Press Enter to try again...";
    cin.get();
}
```

Figure 04.4: Selection

4.5 Control

Control structures are used to control the flow of program execution based on conditions and repetition.

In this system, control structures such as if, while, and continue are used to validate user input and ensure correct program flow.

For example, a while(true) loop is used to repeatedly prompt the user to enter a valid bank name, while if conditions check whether the input is empty or contains valid characters before allowing the program to proceed.

```cpp
paymentMethod = "Online Banking";
string bankName, accountID;

while (true)
{
    cout << "\t\t\t\tEnter Bank Name (Type bb to go back): ";
    getline(cin, bankName);

    if (bankName == "bb")
    {
        system("cls");
        goto SERVICE_MENU;
    }

    if (bankName.empty())
    {
        cout << "\t\t\t\tBank name cannot be empty.\n";
        continue;
    }
    bool hasLetter = false;
    for (char c : bankName)
        if (isalpha(c))
            hasLetter = true;

    if (!hasLetter)
    {
        cout << "\t\t\t\tBank name must contain letters.\n";
        continue;
    }
        break;
}
```

Figure 0.5: Control

4.6 Pointer

Pointers are used to store memory addresses and allow dynamic access to data during program execution.

In this system, pointers are used when interacting with the database, such as handling SQL statements and result sets.

These pointers enable efficient data retrieval and proper memory management when executing database queries.

```cpp
sql::Statement* dstmt = globalCon->createStatement();
sql::ResultSet* dres = dstmt->executeQuery(detailQuery);
```

Figure 0.6: Pointer

4.7 Error Handling

Error handling is implemented to validate user input during service selection.

The system handles non-numeric input using a try-catch block to prevent runtime errors during string-to-integer conversion. It also checks whether the selected option is within the valid range of available services.

If invalid input is detected, an appropriate error message is displayed, and the user is prompted to re-enter the input without terminating the program. This approach improves system reliability and user experience.

```cpp
cout << "\n";
cout << "\t\t\t\t====================== ADD NEW REQUEST =====================\n\n";
cout << "\t\t\t\tAvailable Services:\n";
cout << "\t\t\t\t------------------------------------------------------------\n";

for (int i = 0; i < ids.size(); i++)
    cout << "\t\t\t\t[" << i + 1 << "] " << names[i]
    << "  (RM " << prices[i] << ")\n";

cout << "\t\t\t\t------------------------------------------------------------\n";

cout << "\t\t\t\tSelect your service (1-" << ids.size()
    << ") or (Type bb to go back): ";
getline(cin, option);

if (option == "bb")
    return;

try { choice = stoi(option); }
catch (...)
{
    cout << "\n\t\t\t\tInvalid input.\n";
    cout << "\t\t\t\tPress Enter to try again...";
    cin.get();
    continue;
}

if (choice < 1 || choice >(int)ids.size())
{
    cout << "\n\t\t\t\tInvalid choice.\n";
    cout << "\t\t\t\tPress Enter to try again...";
    cin.get();
    continue;
}

break;
```

Figure 0.7:Error Handling

4.8 Business Rule Implementation

The system implements business rules through program logic. For example, admin registration requires a verification code with a maximum of three attempts. If all attempts fail, access is denied. This rule improves system security and prevents unauthorized admin access.

```
if (accountType == "A") {
    int attempts = 3;
    bool verified = false;

    while (attempts--) {
        cout << "\n";
        cout << "\t\t\t\tEnter Admin Verification Code (Type bb to go back): ";
        getline(cin, verificationCode);

        if (verificationCode == "bb") {

            goto ACCOUNT_TYPE;
        }

        // Correct code
        if (verificationCode == "A12345") {
            verified = true;
            break;
        }


        cout << "\t\t\t\t Invalid verification code. Attempts left: " << attempts << endl;
    }


        // If all attempts used
        if (!verified) {
            cout << "\t\t\t\tVerification failed (3 attempts).\n";
            cout << "\t\t\t\tReturning to Account Registration...\n";
            cin.get();

            return;// Exit the function after 3 failed attempts
        }

}
```

Figure 0.8: Business Roule Implementation

4.9 Complex Calculation Implementation

This system implements a complex calculation to determine the final service price based on customer request history and discount rules.

The calculation begins by retrieving the total number of previous requests made by a specific customer from the database. This value is then used to compute the effective number of requests.

Based on predefined business rules, different discount rates are applied:

- Customers with 3 or more requests receive a 30% discount.
- Customers with 5 or more requests receive a 50% discount.
- Customers with 7 or more requests receive a 70% discount.

The discount amount is calculated by multiplying the base service price by the applicable discount rate, and the final price is obtained by subtracting the discount amount from the base price.
Additionally, a unique request ID is generated dynamically to ensure that each request is uniquely identifiable.

This implementation demonstrates the use of database queries, conditional logic, arithmetic calculations, and dynamic data generation to support business decision-making within the system.

```
//  Calculate Discount
int totalRequests = 0;
string countQuery =
    "SELECT COUNT(*) FROM request WHERE Customer_ID='" + customerID + "'";

stmt = globalCon->createStatement();
res = stmt->executeQuery(countQuery);

if (res->next())
    totalRequests = res->getInt(1);

delete res;
delete stmt;

int effectiveRequests = totalRequests + 1;

double discountRate = 0;
if (effectiveRequests >= 7) discountRate = 0.70;
else if (effectiveRequests >= 5) discountRate = 0.50;
else if (effectiveRequests >= 3) discountRate = 0.30;

double discountAmount = basePrice * discountRate;
double finalPrice = basePrice - discountAmount;

srand(time(0));
string requestID = "R" + to_string(rand() % 90000 + 10000);
```

Figure 0.9: Complex Calculation Implementation

4.10 Report Generation for Analysis Implementation

This function implements the core logic for report generation and analysis in the system.

It retrieves service-related data from the database using SQL aggregation functions such as COUNT and SUM to calculate the number of requests and total service cost.

The function supports dynamic report generation by applying different date conditions, allowing it to be reused for weekly, monthly, and annual reports.

This implementation enables administrators to analyze service performance, monitor demand trends, and support decision-making based on accurate system data.

```
void serviceSalesReport(const string& title,
    const string& dateCondition,
    const string& subtitle = "")
{
    connectToDatabase();
    system("cls");

    cout << "\n";
    cout << "\t\t\t\t =========================================\n";
    cout << "\t           " << title << "\n";
    cout << "\t\t\t\t =========================================\n\n";


    if (!subtitle.empty())
        cout << "\t\t\t\t" << subtitle << "\n\n";

    string query =
        "SELECT r.Service_ID, "
        "        COUNT(*) AS NumRequests, "
        "        SUM(s.Service_Cost) AS TotalCost "
        "FROM request r "
        "JOIN service s ON r.Service_ID = s.Service_ID "
        "WHERE " + dateCondition + " "
        "GROUP BY r.Service_ID";

    sql::Statement* stmt = globalCon->createStatement();
    sql::ResultSet* res = stmt->executeQuery(query);

    bool empty = true;

    cout << "\t\t\t\t             ===Service Report===\n";
    cout << "\t\t\t\t+------------+----------------+----------------+\n";
    cout << "\t\t\t\t| Service ID| Num Requests   | Total Cost(RM) |\n";
    cout << "\t\t\t\t+------------+----------------+----------------+\n";

    double total = 0.0, mn = 1e9, mx = 0.0;
    int count = 0;
```

Figure 0.10: Complex Calculation Implementation

CHAPTER 5: CONCLUSION

5.1 Constraints

i. Authentication and Authorization:

The system implements basic authentication and authorization mechanisms for both administrators and customers. User passwords are securely stored, and administrator access requires an additional verification code with a limited number of login attempts.

The system enforces role-based access control, where administrators and customers are restricted to accessing only the functions relevant to their roles. Customers can view and manage their own requests and personal information, while administrators can manage services, users, and reports.

However, advanced security features such as multi-factor authentication and session-based token management are not implemented due to project scope and time limitations.

ii. Database Integrity:

The database design ensures data integrity through the use of unique primary keys and foreign key relationships between tables. Input validation is applied to prevent invalid data such as empty fields, incorrect formats, and negative values.

Referential integrity is maintained to ensure consistency between related tables such as users, requests, services, payments, and reports. These constraints help reduce data redundancy and improve overall system reliability.

5.2 Future Improvements

i. Add a Secure Payment Gateway:

In future enhancements, the system can be integrated with a secure payment gateway to support real online transactions such as credit cards, debit cards, and online banking services.

This improvement would allow customers to complete payments directly within the system in a safer and more convenient way, while ensuring the protection of sensitive financial information through encrypted and verified payment processes

ii. Improve and Scale the Database System:

The database system can be further optimized and upgraded to support larger datasets and higher system usage. Future improvements may include performance tuning, indexing, backup automation, and migration to a more scalable deployment environment.

These enhancements will help the system manage data more efficiently and improve reliability as the number of users and transactions increases.

iii. Advanced Service & Customer Support Features:

The system can be extended with advanced customer support features such as service history analysis and improved request tracking.

These enhancements would help administrators provide faster responses, better service recommendations, and improve overall customer satisfaction.

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